

BIAS IN DATASET AND AI DECISION-MAKING:

Assessing Dataset Bias, Impacts and Consequences, Exploring Solutions and Future Perspectives

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Abstract

This thesis explores and investigates the source, impact, and mitigation of bias in recruitment datasets using a combination of machine learning, fairness evaluation, and explainable artificial intelligence (XAI) techniques. As AI automated hiring systems become increasingly common, ensuring fairness and transparency in decision-making models is critical. The study begins with identifying bias source using exploratory data analysis (EDA), existing AI toolkit (AIF360, Fairlearn) and XAI within the dataset, followed by implementing baseline classification using five machine learning models: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM)- revealing weaker predictive performance especially for Class 1('Hired') in hiring decisions. To address these issues different mitigation strategies using AIF360 and Fairlearn were applied and result showed Decision Tree responded well in few mitigation strategies to fairness improvements. However, Random Forest model showed consistent highest overall accuracy, precision and recall in baseline and most of the mitigation phases, which showed it's reliability throughout the study. This research aims to contribute to the growing field of fair machine learning in recruitment systems by demonstrating the need for model-specific and context-aware bias mitigation strategies.

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List of Abbreviations

Abbreviation	Meaning
AI	Artificial Intelligence
XAI	Explainable Artificial Intelligence
COMPAS	Correctional Offender Management Profiling for Alternative Sanctions
EDA	Exploratory Data Analysis
LIME	Local Interpretable Model-Agnostic Explanations
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Oversampling Technique
AIF360	AI Fairness 360 Toolkit
DIR	Disparate Impact Remover
DP	Demographic Parity
EO	Equalized Odds
F1-score	Harmonic Mean of Precision and Recall
KNN	K-Nearest Neighbors
LR	Logistic Regression
DT	Decision Tree
RF	Random Forest
SVM	Support Vector Machine
IBM	International Business Machines Corporation

Chapter 1

Introduction

Artificial Intelligence has become an integral part of our modern decision-making processes, playing important role in different sectors, including, healthcare, finance, criminal justice, human resources, resource management, manufacturing etc. The ability of AI to analyze and process large datasets, analyze complex problems, identify patterns and trends, and make predictions efficiently and accurately got widespread acceptance and appreciation in decisions that were taken by human judgements previously. With the help of AI [25] automation, now-a-days complex decision making became easier, time effective, less costly and accurate. However, this growing dependency also creates an alarming concern, which is to make sure that the decision making delivers valid, fair, transparent, and accurate outcomes. Among all those concerns, Bias in Dataset of Decision-making process became a significant challenge to handle. For example, improper Racial profiling is one of the outcomes of Biased Dataset.

Over the past few years, incorporation of AI in every sector was introduced to make system more efficient, scalable and cost-effective, which will lead an estimated 45% of global growth in 2030[27] . From human judgement to AI judgement, this transition influences every sector of our modern life. Now, all sectors are using AI powered decision-making for better predictions and better resource management. This AI decision making outcomes are used to evaluate whether to give loans, or premium insurance or recommending treatments[15]. So, ensuring to use fair, accurate and unbiased Dataset for this AI decision-making process are very crucial for impartial outcomes. Otherwise, it will lead to injustice, inequality or social prejudice.

Bias in Dataset occurs when a certain group or features are over or underrepresented in the dataset, it will lead to unfair and inaccurate outcomes. Dataset bias can be because of improper choice of features, historical imbalance, sampling errors, unequal representation of demographics, class imbalance, favors to sensitive features etc[51]. Sometimes bias in dataset can be a result of inadequate knowledge of data collectors

and their beliefs.

Addressing bias in dataset from the very beginning of the implementations is necessary. Detection bias from the source and mitigation bias in dataset is a necessity as it will improve the AI model performance and will ensure rational decision-thinking [38]. So, its important to have approaches that will address both technical and non-technical strategies to mitigate bias in Dataset. Formulating and ensuring valid balanced Dataset can be one of the technical approaches to mitigate or reduce bias. Ensuring following rules and regulations in respect to use of data and promoting fairness, transparency, accountability is the non-technical approach to address bias in Dataset. The non-technical approach is basically the encouragement and responsibility of an organization to use and collect fair and impartial Dataset in their AI development.

1.1 Motivation

The rapid integration and use of AI decision-making become significant interest point for everyone as it's considered one of the important milestones of technology development. On one hand, it offers the scope of improved, efficient, accurate and scalable system in respect to complex decisions but on the other hand, the issue of Bias in AI powered decision-making outcomes become evident which arises the threat of unfair system. These threats and challenges motivate behind this thesis studies so that the causes and consequences of Bias in dataset of AI powered decision-making can be addressed not only on technical aspects but also with non technical approaches. The major motivation behind these studies are:

- Bias in Dataset creates issue to fairness and transparency in AI driven decisions.
- Mitigating Bias in Dataset will ensure fair and equal treatment for all groups regardless of their gender, ethnicity and demographic characteristics.
- Dataset bias impacts human rights and social justice (for example Job recruitment, HealthCare benefits distribution etc), so eliminating bias from Dataset will ensure fairness in distribution.
- Addressing bias and mitigating it from Dataset will reduce the risk of legal penalties and reputational damage for companies that use large dataset.
- Mitigating bias in Dataset will ensure AI model more FAT (Fairness, Accountability and Transparency) principle, which will increase the public trust in AI driven decision making.

- Reducing bias in Dataset improves the quality of AI model outcomes which makes the model outcomes more reliable, robust and explainable.

One of the key factors of using AI decision-making process is that it gained public trust because of its efficient, trustworthy outcomes [28] but if the system is biased, ultimately people will lose their trust in AI systems. As a result, developing and analysing methods and approaches that will detect and mitigate bias Dataset is essential. The overall goal of this studies will be to understand the effect of different bias and how to prevent or minimize the effect of bias in Dataset so that it AI powered decision-making process can deliver fair and transparent outcomes.

1.2 Research Objectives and Derived Questions

The primary goal of this thesis is to explore, identify, and mitigate bias in dataset in recruitment context. Bias in dataset affect the performance and fairness of AI models, so to assessing and evaluating the existing strategies to detect and mitigate bias is crucial. The study also aims to explore different techniques to identify bias source in the dataset using different data analysis methods, existing AI tools, and use of explainable AI. To understand the impact of bias in dataset, this study will also investigate different bias mitigation techniques and their performance to evaluate their effectiveness. This section outlines the key objectives of this thesis by presenting the key research questions derived from it. All the questions are designed to investigate and understand how bias occurs in a dataset, how it can be identified and mitigated, and how bias-aware data collection will help to handle this situation.

To fulfill the objective of this study, the following research questions have been derived:

Research Question 01

What are the key sources of bias within the dataset in the context of recruitment

This question aims to understand the origin of bias by investigating dataset features, structures, and correlation between features in the recruitment dataset that led to discriminatory results for certain groups.

Research Question 02

How can Bias be detected using different technical techniques?

This question addresses the use of different technical approaches to detect bias using tools such as exploratory data analysis, existing AI tools to detect bias in dataset.

Research Question 03

How do both biased and bias-mitigated dataset affect the model performance and accuracy in recruitment AI models?

This question investigates the impact of model outcomes by comparing the recruitment predictions and explores how mitigation can shift predictive recruitment outcomes.

Research Question 04

How explainable AI (XAI) helps to detect bias within dataset to promote fairness?

This question helps to study the emerging strategies such as Explainable AI that not only detect bias within dataset but also understand why they occur and especially how model is making decisions.

Research Question 05

What data collection strategies can minimize bias?

Finally, this question encourages this study to understand the data collection process as a key source of bias as data collection phase significantly influences model behaviors.

1.3 Thesis Structure

This thesis is organized into six chapters, contributing proper clarification and readability to investigate and mitigate of bias in recruitment dataset. Starting from:

Chapter 01: Introduction

This chapter introduces the topic, states the motivation, describes the objectives and related research questions and finally provides the outline of this study.

Chapter 02: Literature Review and Related Work

This section describes prior work related to bias in dataset, fairness in machine learning, bias in recruitment process, existing bias detection and mitigation techniques using explainable AI also and ends with real-world case scenario to validate the importance of this study.

Chapter 03: Methodology

This chapter Describes the dataset, pre-processing steps, and the technical methods

used for bias detection and mitigation and explains the baseline model to compare.

Chapter 04: Bias Detection

This chapter presents 4 different approaches to detect bias: exploratory data analysis, statistical data analysis, bias identification using existing AI tools and using explainable AI strategies such as SHAP and Lime.

Chapter 05: Bias Mitigation

This chapter resents two different phases of mitigation strategies: (1) addressing target class imbalance using techniques like SMOTE and reweighting, and (2) mitigating feature-level bias using tools from AIF360 and Fairlearn.

Chapter 06: Result Comparison and Analysis

The comparison between baseline model and different model after bias mitigation strategies will be analysed in this chapter.

Chapter 07: Future Work and Conclusion

In this chapter, the limitations of this study will be discussed and summarizing the key findings of this thesis study.

Chapter 2

Literature Review and Related Work

2.1 Background

Dataset bias occurs because of systematic inaccuracies of a machine learning model arising by the issues in its data collection, processing, handling or resulting stages. As a result, most of the time models perform or predict unevenly to its different groups causes discriminatory behaviour. So to ensure a fair, trustworthy and reliable AI powered decision-making system, understanding, managing and addressing dataset bias is very crucial. Numerous studies and research have discovered and identified bias in AI powered decision-making systems. Such studies discuss the impact and reason behind this biasness in the system. According to IBM think 2024 conference, Bias can be found in training data, in the algorithm or in the prediction logics [19]. Caterina Moruzzi and El-Assady highlights the internal causes of biased system can be a result of inappropriate knowledge representation, inadequate explanation, inaccurate or insufficient knowledge and improper intention during AI tools development [11]. Biased Dataset creates systematic discrimination for example in hiring algorithm may favour male candidates over female candidates, recognized university over good GPA etc. Danks and London presented the potential five sources of bias such as Training data, Algorithmic focus, Algorithmic processing, Transfer context and Interpretation bias [9], which need to be addressed to prevent and mitigate bias. Addressing bias is important as it effects systems accuracy, efficiency and fairness. According to Rajiv et al. if the system doesn't produce fair outcomes, it can give unfair predictions for certain groups which can have negative impact on different communities [40].

There are lots of studies and research performed to identify the source of bias in

dataset. According to one the article published by Zilliz, bias in dataset can be introduced by the means of data collection tools and methods as a result historical bias, representation bias, selection bias and measurement bias etc can occur[52]. One of the studies performed by Brown et al. where the authors discuss the importance of studying the protected or critical variables of the dataset which are ‘ race, age, gender, religion, ability/disability, sexual orientation, color, nation of origin, socioeconomic class etc’ at the early data level so that model outcomes do not become discriminatory Brown. According to author Harris, one of the less researched biases compared to other bias is ‘Ageism’ bias[17]. The author also describes that age bias is basically an implicit bias as a result it’s far more challenging to address it. Since in job recruitment sector, job screening and job hiring tools takes basic information of an applicant so these primary words including age plays significant factors in a candidate selection process[32]. For centuries, one of the common stereotypes in recruitment is hiring older workers and taking historical data to train models passes this stereotype in decision-making tools knowingly or unknowingly[22].

To understand how unfairness arise in within a dataset especially AI driven recruitment dataset, it is important to understand common types of bias that occur in dataset and may affect data and decisions of model predictions.

Selection Bias

It occurs when the data collection strategy does not represent of the overall population. In recruitment, this may happen if candidates from certain groups are underrepresented in the dataset[45].

Historical Bias

One of the most crucial bias that need to be addressd is historical bias. It arises when past decisions—such as previous hiring records—reflect societal inequalities for instance, favoring candidates from elite institutions[45].

Measurement Bias

It occurs when features used in the model do not accurately represent the intend to measure. For example, using educational degree can be a proxy variable which can give disadvantage capable candidates without formal qualifications[41].

Proxy Bias

Proxy bias occurs in a dataset when a feature strongly correlates with a sensitive attribute e.g., ZIP code acting as a proxy for race or socioeconomic status[45].

Unbiased and accurate prediction is critical issue of Artificial Intelligence based medical diagnosis. In U.S.A around 10% of this diagnosis was faulty according to a report published by National Academics of Science, Engineering and Medicine [20]. These

misdiagnoses have not only caused problem for the patient but also in resource allocation of hospitals which can be economic consequences of biased AI systems. According to author Adadi and Berrada, all AI- powered recruitment decisions must have the ability to explain its outcomes on the context of trust, fairness and legal standards which can be introduced by the help of XAI techniques [1].

So, preventing and reducing bias in AI decision-making process is much needed study and for years many scholars are trying to find the best possible solutions. Re sampling data, readjusting training data, adjusting algorithms, imposing fairness constraints and creating accountability for fairness can address this problem in the initial phases of development [2]. Several open-source AI tools and libraries have been developed over the year to ensure fair and accurate AI models. Some of them are for detecting bias in dataset, or in model or in the model outcome. FairTest is such an AI tool that detects bias in the dataset by creating associations with the help of checking and mapping between predicted attribute and protected attributes[47]. Another toolkit for analysing dataset is Aequitas, based on the dataset it helps user to decide and identify correct fairness metrics to use for their particular dataset[42]. There are some other tools which assess the outcome of AI based decision-making tool to validate their results against discrimination. Themis is such kind of a toolkit, which generates test suits to evaluate discriminatory outcomes predicted by the predictive model[6]. According to the author Friedler et al. this tool also have fairness comparison library which includes bias detection metrics and mitigation methods for example prejudice remover and also serves as a platform for comparing different fairness algorithms and datasets[14]. Finally AIF 360 first open-source such a tool which combines all this in one package, it deals with bias metrics, bias mitigation algorithms, also explains bias metrics and its practical usability which helps to implement it easily in real-worlds applications[7].

The contribution of this background study in the thesis, is to explore and study the diverse research of the source of bias in dataset, identification of critical and proxy features, detection techniques and exploring mitigation frameworks. This section also explores the study of the existing AI tools to address the bias and discriminatory issues of AI decision-making. The following section will give validation of this contribution with the help of practical implementation and impact assessment with the help of case studies.

2.2 Case Studies

Bias in the dataset is not entirely theoretical, rather it is a very well-documented issue that is addressed with real-world scenarios. The consequences of dataset bias are analyzed by numerous organizations and researchers where they have identified where and when bias occurred and how it creates discriminatory outcomes; in this subsection, few of existing case studies will be presented as concrete evidence of how bias can provide advantages or disadvantages to certain groups. This section will also present a series of scientifically published case studies to illustrate the relevance of these thesis studies, which is bias in the dataset in the recruitment context, and it will also validate a few of the mitigation techniques including Explainable AI. Considering the diverse nature of bias in Artificial Intelligence, the case studies are organized into few groups to discuss, aligning and supporting the core research goal and research question of this thesis.

- First, a few case studies will be discussed where biased data using machine learning models create measurable and questionable advantages or disadvantages outputs to certain groups of people in recruitment and other fields.
- Then cases will be examined, where the primary focus is to address the bias in dataset issue, how to detect and mitigate through technical measures, and discuss the impact of bias dataset aligned with research questions.
- Finally, how Explainable AI(XAI) technology detect and mitigate bias in dataset will be discussed in this section.

2.2.1 Empirical Evidence of Bias-Induced Discriminatory Outcomes in Real-life

Case Study 1

Encoded Bias in Recruitment Algorithms – A practical, intersectional approach to fairness in Machine Learning Algorithms [5]

Background and Context

In 2014, Amazon developed a Resume Screen tool to automate their recruitment system. The tool was trained using their previous historical hiring data. The main function of this tool is to suggest top rank candidates that is like the resumes of their previous successful applicants. It also ranks Biased Behaviour. As the model takes resume as input and based on different feature vector including gender it provides top candidates. It constantly rated high to male applicant then the female applicant.

Source or cause of Bias

This recruitment tool was trained on historical imbalanced and biased dataset as in past the company was male dominant which shows in the dataset also. Another factor was highly reputed university names; the tool always gives more points to the reputed universities rather than qualification. This biased data led to biased outcomes and favors male applicants and reputed university applicants, then the female applicants.

Impact and Mitigation

When the biased behaviour was identified by internal auditing, at first Amazon tried to remove the gender bias but unfortunately, they were unsuccessful to deploy it publicly. As a result, the company decided to shut it down completely rather than taking risk of discriminatory outcomes.

Relevance of Research Questions

This case study is directly connected to most of the research question of this thesis because it illustrates that identifying and collecting proper data is very crucial as without proper handling of critical features led to biased behaviour of a system. A Biased dataset produces discriminatory outcomes. If it's not addressed and mitigate properly and timely, will not only affect the performance of the tool but also raise issues of the company's reputation and fairness.

Case Study 2**Age Bias: A Tremendous Challenge for Algorithms in the Job Candidate Screening Process [18]****Background and Context**

The paper "Age Bias: A Tremendous Challenge for Algorithms in the Job Candidate Screening Process" illustrates in depth analysis of how older job candidates gets unintentionally disadvantages by the automated recruitment tools. The researchers examined and reviewed how common Machine Learning models exhibited age-related discrimination on real-world applicant data.

Biased Behaviour

The author describes that the algorithm consistently favors younger applicants then the older applicants. In some cases, the older applicants even hold equivalent or more qualifications than the younger applicants. Though in some case the dataset doesn't contains the critical feature 'age', but the features like 'date of graduation, employment history' or some other information in resume indirectly served as age indicators as a result the algorithm assigned lower score and ranked older applicants below then younger applicants.

Source or cause of Bias

The root cause of this biasness is because the dataset contains historical data. Features

like graduation year, experience year, employment gaps etc. served indirect indicator of age which are also known as proxy variable. As a result, some of the ML model creates age correlated pattern using those proxy variables even when the feature age is not included in the dataset.

Impact and Mitigation

The study shows that some fairness metrics including disparate impact ratios and equal opportunity metrics were used to measure the bias in the tool. As age bias in recruitment tool violates employment rules and regulations and creates age discrimination, the author states that some adversarial debiasing and dataset balancing can increase partial improvement in the recruitment tool.

Relevance of Research Questions

This case supports the RQ1 and RQ2 as it shows how critical and proxy features contribute and led to a biased system. This study also shows that different mitigation strategies can help to decrease the biasness and discriminatory behaviours.

Case Study 3

Discrimination in Online Ad Delivery: Google ads, black names and white names, racial discrimination, and click advertising [46]

Background and Context

The study titled ‘Discrimination in Online Ad Delivery’ by Lataney Sweeney explored Google’s advertising algorithm which shows racially discriminatory outcomes based on first name associated with specific race. This case study shows how most of the time search engine shows Black African American people photos whenever there is any query about criminal activities.

Biased Behaviour

The author showed that if the ad search or delivery query is about arrest record related, there will be names of black individuals occurs more likely than the white individuals. This discriminated behaviour occurs even when the black individual has no previous criminal records. The reason behind this, is the ad delivery system associates certain race individuals with negative stereotypes regardless of the person’s background.

Source or cause of Bias

The main cause of this biased behavior comes from feedback mechanism from the user, not from the race labels. Over the years, the model learns higher click rates on arrest related ads are connected and associated with black individual names as a result through reinforce learning mechanism, the system always predicts discriminatory outcomes towards black names.

Impact and Mitigation

This ad results influence perceptions of individuals towards black names potentially

affects other people's judgement towards the race without having proper background research. This case raised one of the critical and ethical question of modern era, which is the use of AI for personalization. It also shows using unmonitored AI tool can led to unjustly influence to the society. Though Google publicly didn't acknowledge this discrimination, this case study shows how important to make sure the fairness of the algorithmic advertising.

Relevance of Research Questions

This case study shows how the proxy variables can introduce discrimination in AI systems using feedback mechanisms. It also shows that bias is not only relevant to the recruitment tools but also it affects and influences peoples perceptions.

Case Study 4

Machine Bias: There's software used across the country to predict future criminals. And it's biased against blacks [4]

Background and Context

In 2016 one of the New York based journalistic investigation ProPublica examined a AI system that is widely used in the U.S courts which is COMPAS risk assessment tool. This tool is used by the criminal justice system of U.S to predict the likelihood of criminal reoffending which helps they judge to whether the defendants should be released or detained before trail. Although the system marked as race-neutral, but it shows significant racial bias behavior in its predictions.

Biased Behaviour

The main function of this system is to predict whether the defendant is more likely reoffend by predicting high to low-risk categories. The study found that the system predicts defendant from black race were classified as high risk almost twice then the white defendant. It labelled white defendants as low risk more often than the black defendants where in reality '47.7% white defendant re-offends in compared to 28% of Black reoffend defendant'.

Source or cause of Bias

Although race was not a feature for the COMPAS algorithm but features like prior arrests, age specially location acted as a proxy variable for systematic racial inequalities. As the model was trained on historical data where some location and zip code were known as black neighbourhood, it always showed high risk for those defendants comes from the location.

Impact and Mitigation

ProPublica claimed that they analysed more than 7000 cases in Florida, where they compared algorithms prediction as high risk against actual criminal activities and reveals systematic disparities to black defendants. This investigation sparked debates for

transparency of the system COMPAS though the developers claimed statistical validity of the system as the race was not a direct input.

Relevance of Research Questions

This case study also shows how historical data can serve as a proxy and create structural biases in dataset. It further highlights the necessity of study the critical features which acted as indirect features lead to discriminatory outcomes.

2.2.2 Addressing Bias in Practice: Detection, and Mitigation

Case Study 5

REPAIR: Removing Representation Bias by Dataset Resampling [24]

Background and Context

In Artificial intelligence field Representation Bias means when certain groups or classes are over-represented while others are under-represented. When ML model is trained on dataset where representation bias exists, the outcomes work well for majority group then the minority. The case study ‘REPAIR: Removing Representation Bias by Dataset Resampling’ proposes a systematic approach to address this representation bias in a dataset specially in the training dataset by resampling data which ensures equal representation.

Biased Behaviour

The study shows that most of the time the model performs poorly on minority groups specially with gender or ethnic groups. As a result, it led to discriminatory outcomes and unfair treatment to under-represented subgroups.

Source or cause of Bias

According to the author the main cause this is imbalance data in the dataset which means that unequal subgroups in the dataset specially for feature like age, race, ethnicity etc.

Impact and Mitigation

These results skewed learning outcomes as the model learns from the larger group which results higher error rates with the minority group. REPAIR address these problems by using ‘resampling framework’ which ensures equal training data for each subgroup. The authors claims that traditional oversampling and under sampling techniques may cause information loss where REPAIR maintains diversity with minimum information loss. The framework was tested in synthetic and real-life data and found improved fairness metrics specially for the minority groups.

Relevance of Research Questions

This case study is directly connected with the RQ2 and RQ3. It shows the approaches

to minimize biasness using techniques like REPAIR. It also illustrates that model performance and fairness can be improved without compromising model validity.

Case Study 6

Using Explainability for Bias Mitigation: A Case Study for Fair Recruitment Assessment [44]

Background and Context

The author of this study developed a bias mitigation algorithm named ‘dubbed Proxy-Mute’ with the help of explainable AI(XAI) techniques to detect and mitigate dataset bias in recruitment algorithm. The author focused on improving the performance of the algorithm by examining real-world applicant screening dataset and discusses systematic approach using explainable AI.

Biased Behaviour

The study revealed that resume screening model often rely on variables like university name, job title, employment gaps with the critical features like race and gender. These features effect the outcomes by learning social patterns and stereotypes resulting unfair scoring to certain groups.

Source or cause of Bias

According to the author, most of the dataset bias came from imbalance historical data, when the model used corelated features and predict injustice outcomes even though protected attributes were not defined.

Impact and Mitigation

One of the XAI approaches to handle this kind of situation is using SHAP values, which is basically detecting which features contributes disproportionately influenced to biased decisions by analysing model predictions using feature importance. Once identifying the high impact biased features, the author suggests a debiasing approach to remove those features and model retraining which shows more equitable precision across groups.

Relevance of Research Questions

This case study is related to the RQ5, as it shows how XAI techniques can detect and mitigate bias specially hidden bias from dataset and improve the performance of the model.

Case Study 7

IDENTIFICATION AND MITIGATION OF BIAS USING EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI) FOR BRAIN STROKE PREDICTION [30]

Background and Context

The authors Mohammad and George studies and explores the role of XAI in detecting and mitigating biased outcomes in medical predictions. According to the author using XAI techniques can improve the accuracy and model performance in healthcare system which is crucial for practitioners for their better diagnosis.

Biased Behaviour

The author describes that the model always predicts non-stroke outcome accurately then stroke outcomes. So the researcher used XAI LIME and SHAP techniques to understand why the model made its decision on granular level by analysing the feature importance.

Source or cause of Bias

After analysing, it follows that the dataset has imbalance data, where non-stroke patient data was more than stroke patient data. As a result, the model couldn't predict any true positive outcomes for the stroke and gives result 0 for all precision, accuracy and recall.

Impact and Mitigation

After learning the outcomes, the researcher used Both LIME and SHAP techniques in 2 steps to improve the performance. For first step the research performed oversampling to both stroke and non-stroke classes which improves their f1 score to 81.78%. In the second phase they used LIME techniques to analyse the hidden relationships of features specially identifying which features has most influence on the predictions. After that the author states that they found feature like 'age, heart disease' influence the outcomes a lot. So, to reduce this over-dependency, the authors again did a balancing class with the feature 'heart disease' and found the model performed better than last time with 83.34%.

Relevance of Research Questions

This case highlights importance of XAI in automated decision making and validate RQ5 for this thesis purpose.

Chapter 3

Methodology

This chapter outlines the systematic approach employed to explore bias in recruitment dataset and investigate its influence on machine learning model-based AI-hiring decisions. The methodology is structured and designed to align with the research goals and objectives using exploratory data analysis, statistical data analysis, AI tools for detecting and mitigating bias and explainable AI (XAI) methods. The methodology aimed to ensure a comprehensive analysis of dataset bias using a synthetic recruitment dataset containing 1500 candidate's observation.

3.1 Research Design

3.1.1 Systematic Literature Review (SLR) for RQ1 and RQ5

The systematic literature review gives the opportunity to explore existing research on the purpose of bias in dataset in the recruitment sector. SLR techniques are used to address RQ1 and RQ5 – which focus on finding the key attributes that causes Bias in dataset(RQ1) and identifying approaches for Bias-aware data collection strategies(RQ5).The goal of this review and investigation is to find the root causes of Bias in dataset in recruitment context which includes examining the critical features such as ‘gender, age etc’. Another aim for this review is to find out best practices for data collection strategies so that it can minimize the impact of discriminatory outcomes of a biased Dataset.

Search Strategy

The search strategy employed in this study to identify relevant scientific published studies is summarized in table 3.1. This includes the name of the search databases, the keyword used or applied during the search process.

Databases Searched	Keywords
ACM Digital Library, IEEE xplore, ScienceDirect, Google Scholar and arXiv(repository)	Bias in Dataset, Bias in recruitment Dataset, Critical attributes in Dataset, Protected and proxy attributes in Dataset, Discriminatory outcomes in ML , data fairness, ethical AI, Data Collection Strategies and frameworks

Table 3.1: Search Strategies

Search Query

A detailed summary of the search queries used in different databases is presented in table 3.2. These queries were carefully formatted, designed, analyzed and tested to capture the relevant literature on the dataset bias in recruitment system.

Search Query ID	Related Research Question	Keywords Used
SQ1	RQ1: Sources of Dataset Bias	"dataset bias" AND ("sensitive attributes" OR "protected attributes" OR 'proxy attributes')
SQ2	RQ1: Types of dataset bias in recruitment data	"recruitment dataset bias" AND ("hiring discrimination" OR "gender bias" OR 'Age bias' OR 'education bias')
SQ3	RQ5: Best practices during data collection	"fair data collection" OR "data documentation standards"
SQ4	RQ5: Guidelines for bias-aware dataset design	"best practices for data collection" OR "bias-aware frameworks in AI datasets"

Table 3.2: Search Queries and Keywords

These search queries were executed on IEEE Xplore , ACM digital Library and Google Scholar to test the efficiency of these queries. The main reason for these queries is to find relevant peer-reviewed publications on bias in the dataset.

3.1.2 Empirical Design for RQ2, RQ3 and RQ4

The second phase of the research design section is to test and validate experimental finding from SLR search query by applying machine learning models, bias mitigation AI tools and Explainable AI techniques. This subsection addresses RQ2, RQ3 and RQ4 and outlines the workflow of the experimental design to achieve the goals of the stating research question, which will be implemented in later section and will be discussed in detail.

3.2 Dataset Description

This section provides a comprehensive overview of the used dataset in this research. It also provides detailed information of the source, structure and attributes relevant to bias in dataset in recruitment context.

3.2.1 Data Source

The dataset used in this study is a synthetic recruitment data designed to analyse bias in dataset for recruitment context. Each row represents a unique applicant with a column describing their personal information and the result of initial assessments. According to the source of the dataset, this dataset is available in publicly format so that other researchers can study it for machine learning prediction and bias-aware machine learning experiments. The goal of selecting this data set is to ensure the basic demographic diversity that is required to analyze bias in the data set and the hiring decision in a recruitment scenario.

Source: The data is sourced from Kaggle’s ‘Predicting Hiring Decisions in Recruitment Data’ repository¹

This Dataset is publicly available under the CC BY 4.0 license which allows anyone to use it for study purpose. This Dataset contains 1500 observations and 11 features, which makes it a standard dataset for analysing recruitment tools.

3.2.2 Feature Overview

The dataset contains candidate information; each records represent various attributes that affect the hiring decision process in the applicant recruitment system. The dataset contains 1500 candidate observations with 11 features. It contains both demographic and initial assessment related features of candidates, and the attributes can be categorized as follows:

Demographic Features

These includes the feature ‘Age’ (integer values in the range 20 to 50) and ‘Gender’ (Binary encoded, Male (0), female (1)), which commonly represents as protected attributes under anti-discrimination law, as a result they are critical for bias detection and mitigation analysis.

¹Dataset Source: <https://www.kaggle.com/datasets/rabieelkharoua/predicting-hiring-decisions-in-recruitment-data> (Accessed: June 18, 2025).

Qualification and Professional Features

Features like ‘EducationLevel’, ‘ExperienceYear’ and ‘PreviousCompanies’ indicates the applicant’s educational qualification and past working experience. These features can be critical or proxy features for model prediction and potentially introduce bias for example- previous number of companies can reflect the stability of a applicant career.

Geographic Feature

‘DistanceFromCompany’ captures the physical proximity of a applicant to the work-place. This geographic distance may potentially introduce indirect or proxy bias for example candidate who lives far away from the company gets rejection due to indirect distance discrimination based on the proximity factor.

Assessment Features

Features like ‘InterviewScore’, ‘SkillScore’ and ‘PersonalityScore’ represents the initial assessment results of a candidate and can be considered as critical features within dataset. These features are very crucial for hiring decision. As these features are among the key feature for recruitment purposes, must be analysed further for ensuring no correlation with bias within dataset.

Contextual Feature

‘RecruitmentStrategy’ feature indicate the recruitment channel used to attract candidates. Analysing this feature is important as it may create procedural bias if not addressed.

Target Feature

‘HiringDecision’ is the target attributes within the dataset in the binary form as class ‘Hired (1)’ -Hired candidates: 465 and ‘Not Hired (0)’-Not Hired candidates: 1035.

3.2.3 Feature Categorization for Bias Analysis

In the context of bias in the dataset and bias analysis in dataset, certain attributes or features are considered critical or sensitive or protected, as these features can potentially lead to direct or indirect discrimination in an AI model prediction. So, it is essential to identify and prioritize these critical features and their influence on outcomes to design bias-free AI decision-making tools, especially in hiring decision.

Protected Feature (2 features)

According to the author, Yarger et al. 'Gender and Age' are considered protected attributes in most of the recruitment context[50] as there are many laws to ensure it, for example German general Equal Treatment Act ensures equal employment opportunities at workplace in Germany [12].

Sensitive Feature (3 features)

EducationLevel, DistanceFromCompany and RecruitmentStrategy features may correlate with indirect discrimination and introduce societal inequalities.

Critical Feature(5)

Assessment features (InterviewScore, SkillScore, PersonalityScore), Experienceyears and PreviousCompanies features are most relevant features in consideration of hiring decision so without proper analysis and evaluation may create indirect bias.

Fairness metric evaluation, correlation analysis, etc. between features were performed in later sections to illustrate the influence and affect of these features in recruitment decision.

3.2.4 Data Format and Encoding

The dataset contains features in three formats as shown in Figure-3.1 are integer, binary and float. Among all the features Gender and target variable are in binary, where distance from company feature is in float values. As all the features in numerical values, no need to encode any feature values.

```
Binary columns: ['Gender', 'HiringDecision']
Integer columns: ['Age', 'EducationLevel', 'ExperienceYears', 'PreviousCompanies', 'InterviewScore', 'SkillScore', 'PersonalityScore', 'RecruitmentStrategy']
Float columns: ['DistanceFromCompany']
```

Figure 3.1: Features data types

3.2.5 Dataset Scope

The dataset used in this study in the size of 1500 records which ensures adequate analysis validity for detecting bias presents in the dataset specially for demographic sub-groups. The dataset contains 11 features, among these 10 attributes serve as an input attribute for bias detection, mitigation, and prediction in recruitment context and are used throughout the thesis. These various features enable the opportunity for the analysis of direct bias (using protected or critical attributes) and indirect bias (using proxy variables) within the recruitment dataset.

3.3 Data Pre-processing

Effective data pre-processing is a crucial and critical phase in data analysis pipeline for improving data quality and ensuring machine learning model readiness and fairness. In the context of recruitment dataset data quality, comparability, interpretability, and consistency across features significantly influence the outcomes. So preprocessing acts as a bridge between raw recruitment data and structured, analyzable input format for better statistical analysis and model prediction. This section discusses few pre-processing steps to understand the dataset for example cleaning, handling missing value, duplicate rows and scaling of numerical features. For the initial analysis and to get a comprehensive overview, few of the Python library was used to reveal the dataset information and preprocessing techniques.

3.3.1 Data Quality Assessment

Initial data quality ensures relevance of the features, completeness and distributional balance of all 11 features across 1500 observations in recruitment context. In this study data quality assessment was performed to find the completeness of features using missing value checking method. Particular attention was paid to find any inconsistency in the entries of the dataset. Multi-dimensional features assessment helped to validate the machine learning models and their outcomes.

3.3.2 Dataset Information

Using ‘.info()’ function, the number of observations, number of features, data types and no-null entries were revealed to identify mismatching values, as shown in the Figure-3.2.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1500 entries, 0 to 1499
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Age                                   1500 non-null   int64
1   Gender                               1500 non-null   int64
2   EducationLevel                       1500 non-null   int64
3   ExperienceYears                      1500 non-null   int64
4   PreviousCompanies                   1500 non-null   int64
5   DistanceFromCompany                 1500 non-null   float64
6   InterviewScore                      1500 non-null   int64
7   SkillScore                          1500 non-null   int64
8   PersonalityScore                    1500 non-null   int64
9   RecruitmentStrategy                 1500 non-null   int64
10  HiringDecision                      1500 non-null   int64
dtypes: float64(1), int64(10)
memory usage: 129.0 KB
None
```

Figure 3.2: Basic information of the data

3.3.3 Data Cleaning

In this subsection, missing values and duplicate entries were addressed because in recruitment data is a synthetic dataset so it can contain incomplete information or records. The goal of this subsection is to minimize the effect of inconsistent and incomplete records and their influence in model performance. However, in the dataset there was no categorical and numerical features with missing and duplicate values were found, which is shown in Figure- 3.3, makes the dataset an ideal dataset in the context of recruitment.

```
Missing Values Per Column:
Age      0
Gender   0
EducationLevel  0
ExperienceYears  0
PreviousCompanies  0
DistanceFromCompany  0
InterviewScore  0
SkillScore  0
PersonalityScore  0
RecruitmentStrategy  0
HiringDecision  0
dtype: int64

Number of duplicate rows: 0
```

Figure 3.3: No missing and Duplicate values were found

3.3.4 Outlier Detection

In the context of recruitment, outlier detection is a significant pre-processing step as outliers or anomalies can distort or skew statistical analysis, influence ML models negatively as a result models may predicts biased outcomes. In recruitment context, anomalies can be a result of inaccurate data entry, extreme high or low values in features. To analyze the impact and influence of anomalies in the model, outlier detection technique was performed in both visual and statistical methods shown consistently in Figure- 3.4 and Figure-3.5, which are consecutive box plot and Interquartile Range (IQR). After the implementation, no anomalies were found in the dataset.

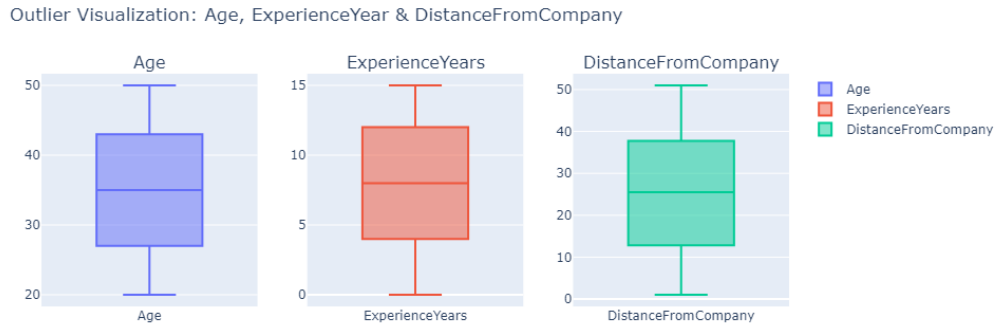


Figure 3.4: Outlier detection for basic information features

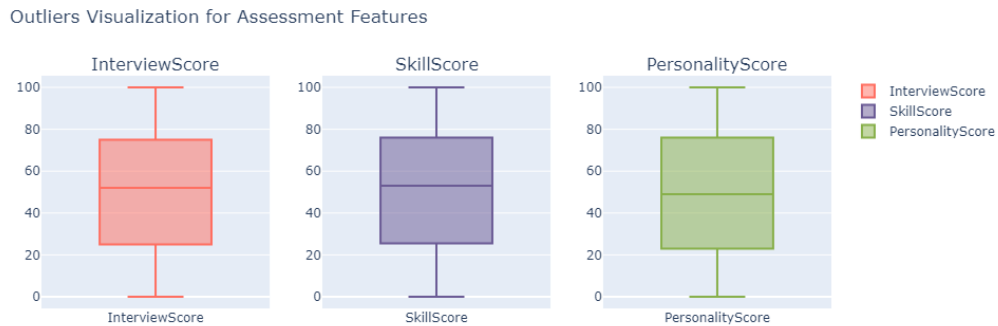


Figure 3.5: Outlier detection for assessment features

3.3.5 Feature Transformation

Feature transformation aims to reduce the skewness and tries to improve the distribution of numerical features of a dataset. The goal of feature transformation is to minimize the negative effect of skewness, and by doing so it enhances the model performance. Some machine learning models like logistic regression, SVM and KNN are sensitive towards normality. So, in this analysis, the numerical feature distribution was visualized as shown in Figure-3.6, which shows the requirement of feature transformation as the different features have different scaling.

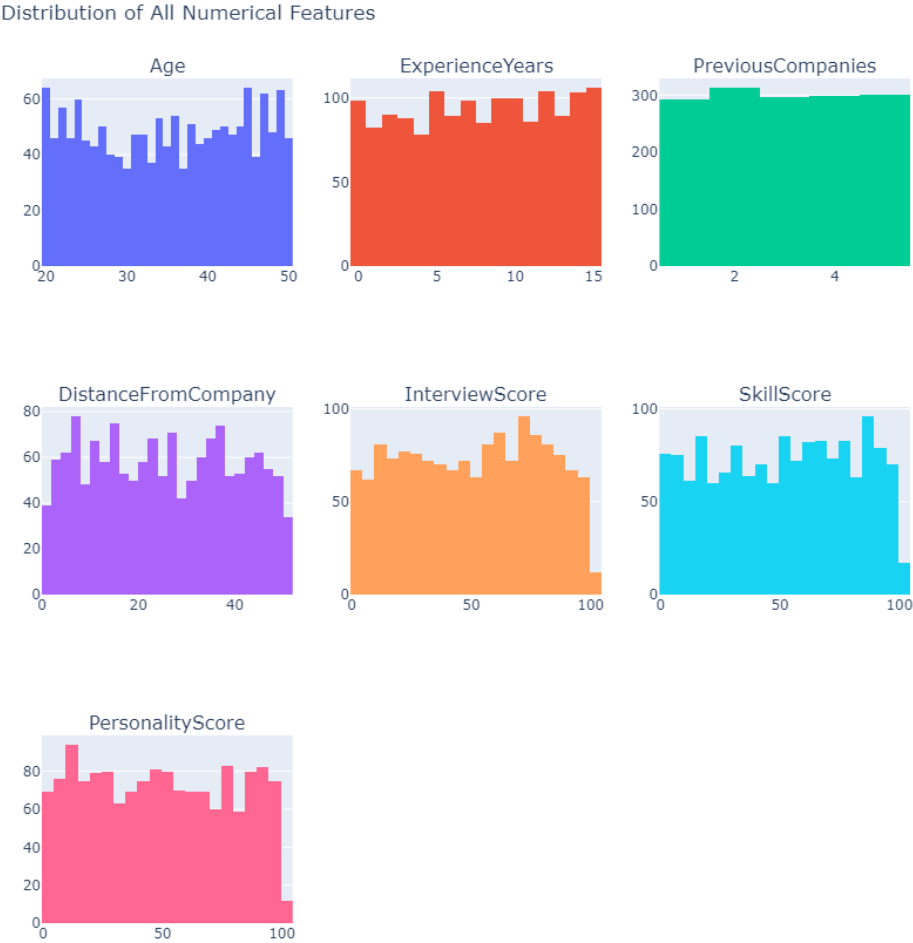


Figure 3.6: Distribution for all numerical features

To ensure whether transformation is really needed, this study also performed the 'Shapiro-Wilk test' method as shown in the to see the p-value, shown in Table-?? for numerical characteristics, which shows a very low P-value which indicates non-normality.

Feature	P-value	Analysis
Age	1.28×10^{-23}	Not normal
ExperienceYears	4.49×10^{-23}	Not normal
PreviousCompanies	1.40×10^{-31}	Not normal
DistanceFromCompany	1.15×10^{-21}	Not normal
InterviewScore	6.48×10^{-21}	Not normal
SkillScore	1.03×10^{-21}	Not normal
PersonalityScore	4.78×10^{-22}	Not normal

Table 3.3: Normality Test Results for Key Features

To improve the prediction of such models that are sensitive towards skewness, the Yeo-Johnson power-transformation technique was implemented which shows in Figure-3.7improvement towards normality.

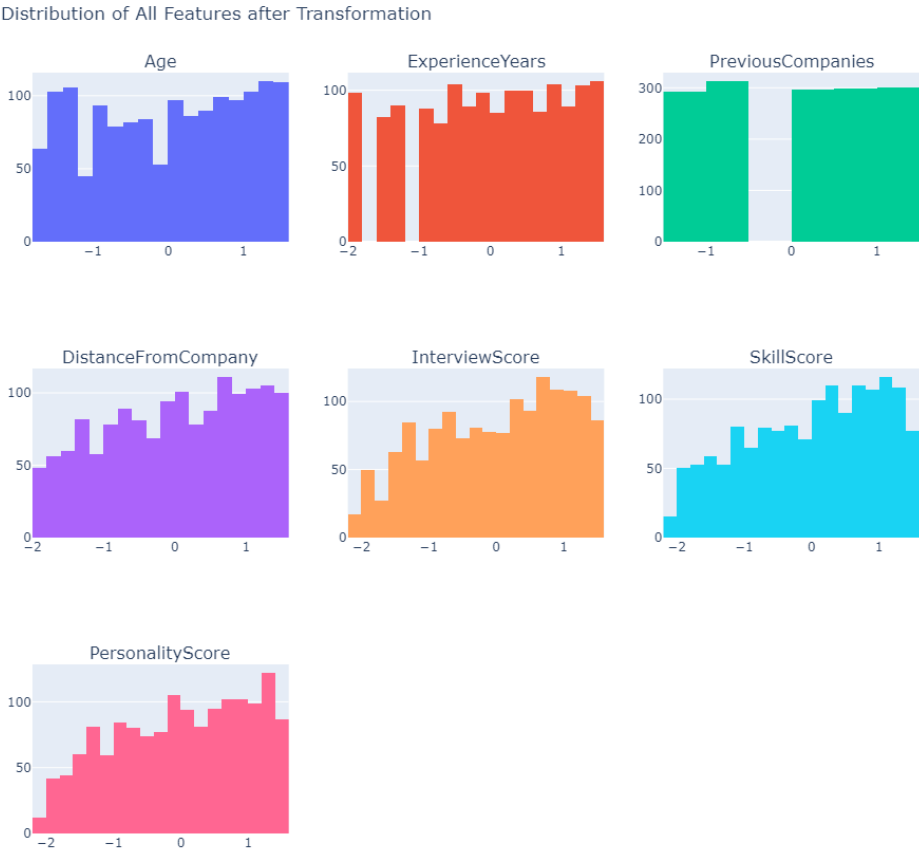


Figure 3.7: Distribution for all numerical features after Transformation

By comparing both p-values in Figure-3.8, it validates the effectiveness of implementing transformation to normalize data.

Feature	p-value (Before)	p-value (After)
Age	1.29e-23	1.13e-23
ExperienceYears	4.49e-23	3.10e-23
PreviousCompanies	1.40e-31	1.17e-31
DistanceFromCompany	1.15e-21	4.23e-21
InterviewScore	6.48e-21	7.68e-21
SkillScore	1.03e-21	8.36e-22
PersonalityScore	4.78e-22	5.26e-21

Figure 3.8: P-Value Comparison: Pre and Post-Transformation

3.4 Evaluation Metrics

3.4.1 Fairness Metrics

Demographic Parity Metrics[33]

Another name for Demographic Parity Metrics is statistical parity. This metric measures if the output of a model is dependent on a feature or not. For example, whether the model prediction of hiring is gender dependent.

Disparate Impact Ratio Metrics(DIR)

It measures the disproportionate outcomes between privileged and unprivileged groups.

$$\text{DIR} = \frac{\text{Hiring Rate of Unprivileged Group}}{\text{Hiring Rate of Privileged Group}} \quad (3.1)$$

If $\text{DIR} = 1$, both groups are treated equally.

If $\text{DIR} < 1$ the unprivileged group hiring rate is less which causes potential bias.

Equal Opportunity Metrics

This evaluation metric is used to judge the fairness of a model. So the aim for this evaluation matrix is to find if the model prediction is fair across different groups, it measures this by focusing on the true positive rate for the positive group. By this, the metric ensures that the group predicted to be positive should be in the positive domain.

3.4.2 ML Model Performance Metrics

These metrics evaluate the ML models' outcome based on their effectiveness and correct hiring predictions. The evaluation shows how well a machine learning model can predict.

Accuracy

One of the fundamental evaluation matrices is accuracy. Accuracy increases as a model increases the amount of correct predictions. So this evaluation sees the amount of correctly predicted values as a proportion of the total number of values.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3.2)$$

Here,

TP = True Positive,

TN = True Negative,

FP = False Positive,

and FN = False Negative.

Precision

Precision is also a very popular evaluation metric for machine learning models, which is especially used in classification problems where the cost of a false positive is high. Precision is the ratio of accurately predicted positive observations (True Positives) to all predicted positives (True Positives + False Positives). So in other words, precision looks at the values which are predicted as true and were supposed to be in the true domain, the equation for precision is,

$$\text{Precision} = \frac{TP}{TP + FP} \quad (3.3)$$

Here,

TP = True Positive,

FP = False Positive

Recall

Recall is also another popular evaluation metric used to evaluate machine learning models. Recall is calculated based on the actual positive cases, which are predicted to be true in proportion to the total correct prediction. The equation for recall is,

$$\text{Recall} = \frac{TP}{TP + FN} \quad (3.4)$$

Here,

TP = True Positive,

FN = False Negative

F1-score

F1-score is an evaluation metric that actually exposes the balance of precision and recall. In other words, F1-score is the harmonic mean of precision and recall. The equation is,

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (3.5)$$

3.4.3 Explainable AI Metrics

For Shapley Additive Explanations (SHAP):

Shapley additive explanations explain the machine learning model based on the prediction generated by the model.

Global SHAP Summary

The Global SHAP (SHapley Additive ExPlanations) summary plot gives a high-level overview of the machine learning model by aggregating the average influence of each

feature on the model's output across all predictions. This global perspective assists in identifying the most influential traits and potential sources of bias in the recruitment decision-making process.

SHAP Feature Importance

SHAP feature importance shows the most important feature for the model, and exposes the root feature for bias. It is determined by having a closer look at the model outcome with the influence of different features. The SHAP value for a feature is defined as

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (3.6)$$

SHAP feature importance calculates how much each input feature adds to a model's predictions using SHAP values based on Shapley values from game theory. SHAP assigns a value to each feature for each prediction, indicating whether it improves or degrades the prediction's performance in comparison to the model's average output. This method gives a fair, consistent, and understandable measure of feature impact, which is beneficial for both model transparency and bias identification.

SHAP Force Plot Local Interpretation

This metric provides a visualization based on which it can be determined which feature pushes the model's decision most and which one the least. In the plot, the features that influence the most are shown in red and the ones that contribute less are shown in blue. Each push's size correlates to the magnitude of the feature's SHAP value .

For Local Interpretable Model-agnostic Explanations (LIME):

LIME interprets the prediction of a model by approximating it locally with an interpretable model; the main focus of this technique is looking at the explanation of a prediction for a single instance rather than analyzing the model's behavior globally.

Global Feature Importance

Based on each feature's contribution towards a model's prediction, global feature importance is measured. The process involves evaluating the impact of each feature on the output, typically by averaging its contribution over all data instances. To get global feature importance from LIME, the average absolute weight of each feature across all locally trained interpretable models is obtained:

$$\text{Importance}_i = \frac{1}{n} \sum_{j=1}^n |w_i^{(j)}| \quad (3.7)$$

Here,

- Importance_{*i*} is the estimated global importance of feature *i*,
- $w_i^{(j)}$ is the weight (coefficient) assigned to feature *i* by the locally fitted interpretable model for instance *j*,
- *n* is the total number of instances for which local explanations are generated.

Group-level Disparity

This technique refers to any unequal treatment for any demographic or protected groups in a model's prediction. It shows if the model is showing any favor towards a group. This plays a vital role in finding a bias, which later on helps to mitigate it.

Local Explanation

Local explanation focuses on individual instances and how much each feature contributes to that prediction. Unlike global explanation, which works on the entire dataset, local explanation works on each decision.

3.5 Baseline Model

3.5.1 Experimental setup

Environment and toolkit:

All the experiments were implemented using Python in Jupyter Notebook using following libraries and tools:

- Pandas and NumPy for data loading and manipulation.
- Matplotlib, Seaborn for data visualization.
- Scikit-learn for pre-processing, data splitting, model development and model evaluation.
- IBM AIF360, Fairlearn for fairness evaluation and bias mitigation (phase 2).
- LIME and SHAP techniques from explainable AI for model interpretability (phase 3).

Dataset and Data Splitting:

The dataset used for this study was describe in methodology chapter 'Recruitment data' of various candidates with features such as Age, Gender, Education level, Experience Years, Interview score etc. and target variable as Hiring decision feature.

For evaluation purpose, dataset was divided into testing and training sets, the split was into 80% training and 20% for testing and the split ratio was maintained throughout all the phases of the experiment.

In this preliminary phase, multiple classification machine learning model were built and implemented to create a baseline understanding for evaluating predictive fairness and performance metrics. The objective of this phase is to assess how traditional machine learning models perform predictions in recruitment dataset without applying bias mitigating and fairness-aware strategies.

3.5.2 Models Implemented

The following machine learning model were implemented based on the finding of literature review and their use for classification problem[31]. As this section focuses only the implementation of machine learning models and their result evaluation, the information and process of handling missing and duplicate data, data splitting and evaluation criteria [37] are provided in details in the Section: 3 Methodology chapter.

3.5.2.1 Logistic Regression

The first model that was implemented is Logistic Regression as it achieves good balance of accuracy and interpretability for binary classification problem. The model was trained on the full set of features during training following standard pre-processing steps. As the values of categorical features is either integer or binary, the main pre-processing step was achieved for scaling the numerical features. In this phase, no bias mitigation and feature selection techniques were incorporated during implementation.

Model Performance

```

Accuracy: 0.87
Confusion Matrix:
[[198  17]
 [ 23  62]]
Classification Report:
              precision    recall  f1-score
0               0.90         0.92         0.91
1               0.78         0.73         0.76

```

Figure 3.9: Evaluation results of Logistic Regression

The model achieved 87% accuracy, shown in Figure-3.9 in the predictions on the

test set ,however the model showed lower precision and recall to its minority class which is 'Hired' class in 'HiringDecision' feature(target feature).

3.5.2.2 Decision Tree

The Decision Tree classifier was implemented to identify if there is any pattern in candidate selection decision will all the features (including sensitive, influential feature) with default parameter. The Decision Tree model is one of the most useful models in the baseline phase as it works well on both numerical and categorical features.

Model Performance

```
Accuracy: 0.88
Confusion Matrix:
[[192  23]
 [ 13  72]]
Classification Report:
              precision    recall  f1-score
0               0.94         0.89         0.91
1               0.76         0.85         0.80
```

Figure 3.10: Evaluation results of Decision Tree

The model achieved 88% of accuracy as shown Figure3.10with improved recall in minority 'Hired' class.

Visualization of Decision Tree: The decision tree in figure-3.11, shows which features contribute most significantly in the hiring decisions.

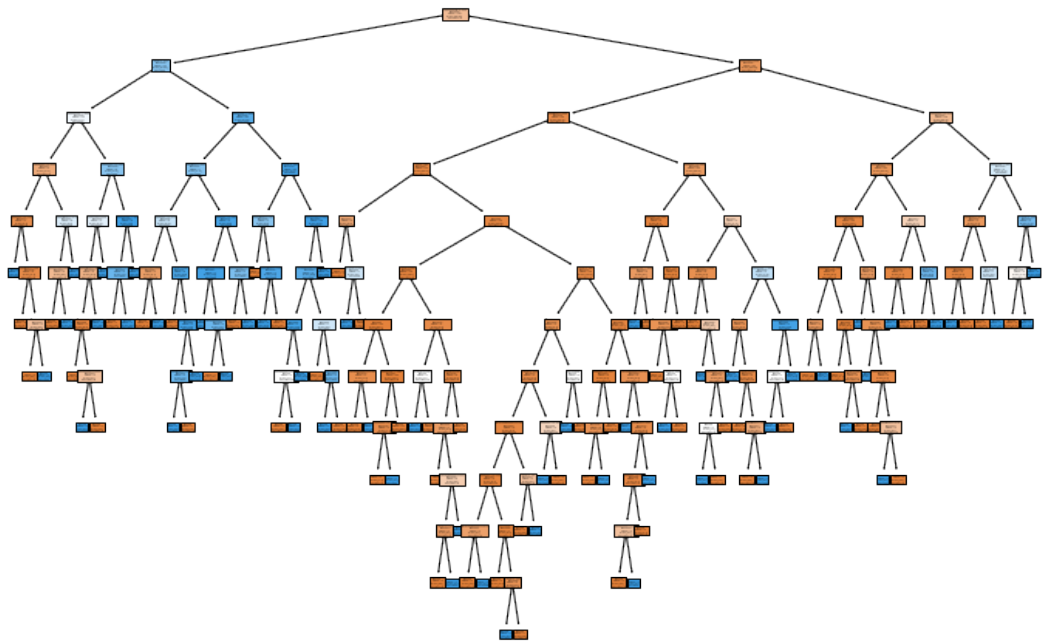


Figure 3.11: Decision Tree

One of the advantages of using Decision Tree classifier is it provides feature importance by generating the decision tree. As shown above, Figure- 3.12 shows the most influential features, among them RecruitmentStrategy contributes high in hiring decision.

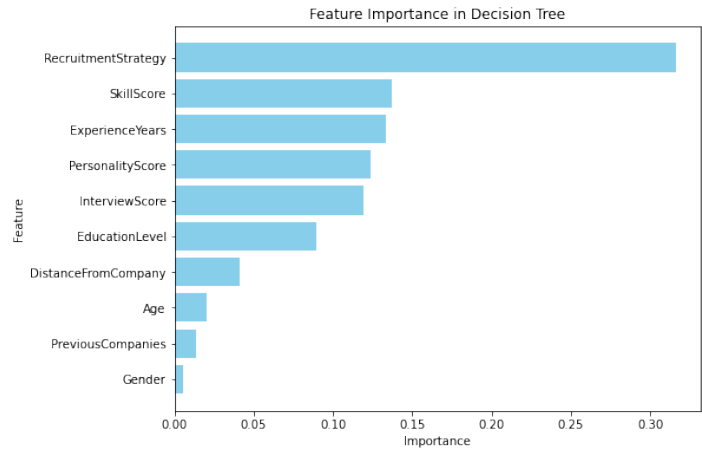


Figure 3.12: Decision Tree Feature Importance

3.5.2.3 Random Forest

Random Forest classifier was implemented in this phase as it combines multiple decision tree with default parameters and aggregate their predictions to improve the decision

accuracy. This model was implemented to increase the robustness and to address the if there is any complex pattern in decision making.

Model Performance

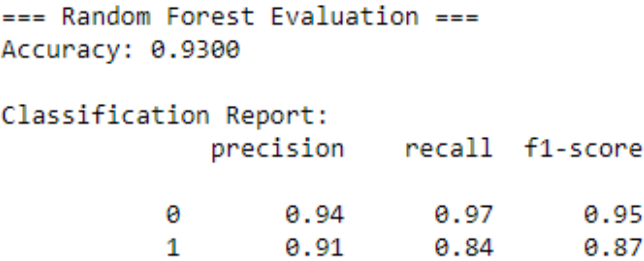


Figure 3.13: Evaluation results of Random Forest

The random forest model has achieved the highest accuracy to date, which is 93% shown in Figure-3.13. It showed a highly strong performance for the majority class and improved the precision in a significant way for the minority class, indicating a balance prediction.

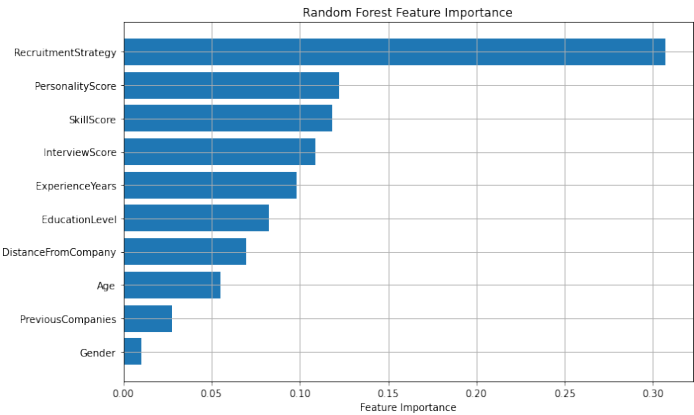


Figure 3.14: Random Forest Feature Importance

As shown in Figure- 3.14, the RecruitmentStrategy feature is the most influential feature, however, it showed PersonalityScore as the second important feature. This specific feature did not rank highly during any exploratory and statistical analysis, suggesting that the random forest may capture the relationship of personality score and hiring decision that EDA and statistical analysis may overlook.

3.5.2.4 K-Nearest Neighbors(KNN)

The K-Nearest Neighbor classifier was implemented to make predictions as it is a distance-based algorithm that assigns a data point-based majority class to their near-

est neighbors. As KNN model is highly sensitive to scaling, imbalanced data, the basic pre-processing techniques were performed before implementing the model.

K-value for KNN

To identify the best K-value that is highly crucial for model performance, a range of K-values was used using cross-validation techniques, shown in Figure-3.15 to select the optimal K-value, which is K= 13.

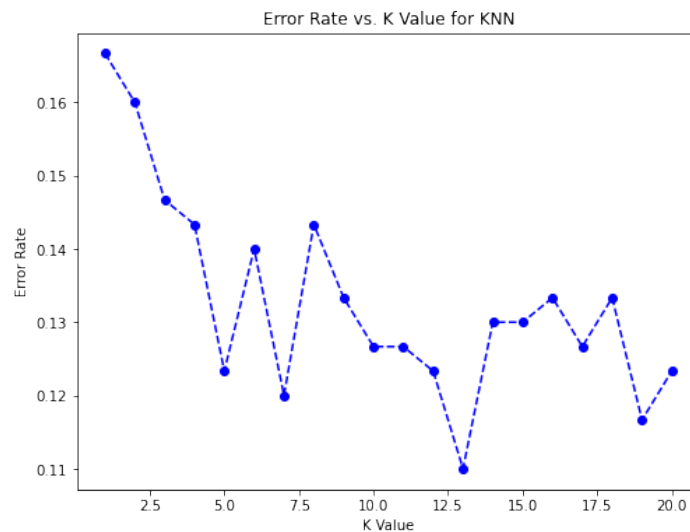


Figure 3.15: Optimal K-value

Model Performance

```
Accuracy: 0.89
Confusion Matrix:
[[208  7]
 [ 26 59]]
Classification Report:
              precision    recall  f1-score
0               0.89         0.97         0.93
1               0.89         0.69         0.78
```

Figure 3.16: Evaluation results of KNN

Using the optimal K-value($k=13$), the model accuracy was 89%, shown in Figure-3.16. As KNN is sensitive to imbalance class, the model struggles to predict positive hiring which shows low recall in minority class.

3.5.2.5 Support Vector Machine(SVM)

The Support Vector Machine classifier was implanted using linear kernel (using SVC class) as its one of the powerful supervised classification algorithms. Since SVM also

sensitive to scaling so all numerical values were standardized during pre-processing phase.

Model Performance

```

Accuracy: 0.90
Confusion Matrix:
[[204  11]
 [ 19  66]]
Classification Report:
              precision    recall  f1-score

0               0.91         0.95         0.93
1               0.86         0.78         0.81

```

Figure 3.17: Evaluation results of SVM

The model predicts 90% accuracy with strong, accurate performance on the majority class, as shown in Figure-3.17. For minority class the recall value was slightly low then precision which can be the reflect of target variable class imbalance.

Result and Key Observation Overall accuracy and performances metrics (class-wise performance) of the five implemented models are shown in the following Figure-3.18 for a balanced view of model predictions and effectiveness.

Model Name	Accuracy	Class	Precision	Recall	F1-score
Logistic Regression	0.87	0	0.90	0.92	0.91
		1	0.78	0.73	0.76
Decision Tree	0.88	0	0.94	0.89	0.91
		1	0.76	0.85	0.80
Random Forest	0.93	0	0.94	0.97	0.95
		1	0.91	0.84	0.87
KNN	0.89	0	0.89	0.97	0.93
		1	0.89	0.69	0.78
SVM	0.90	0	0.91	0.95	0.93
		1	0.86	0.78	0.81

Figure 3.18: Class-wise Result Analysis for baseline model

- All models performed well , Random Forest achieved highest accuracy with 93% and SVM with 90%.

- Across all model, Class 0 (which is ‘ Not Hired’) constantly showed higher precision, recall and F1-score.
- The disparity between Class 0(Not Hired), Class 1(Hired) indicates that the baseline model are generally confident to predicting candidates who should not be hired.
- As a result, in recruitment scenario, it will create a potential bias towards positive predictions, means it will lead to a decision where qualified candidates will get rejected.

The findings from baseline models highlights the importance of applying the bias mitigation techniques so that the model can also predict positive hiring prediction well. During Random Forest and Decision Tree implementation, it showed the feature importance and through both model it was revealed that ‘RecruitmentStrategy’ , is the most influential in hiring decision. There are other features also have notable influence whereas demographic features like ‘Gender’ and ‘Age’ had relatively low importance. In the later phase, through bias mitigation and fairness analysis, all these features will be analysed for more balanced and accurate decision-making.

Chapter 4

Bias Detection

This chapter presents a holistic bias detection framework using both technical and non-technical approaches of the recruitment dataset. The goal of this chapter is to find and identify any bias pattern using multiple detection framework across 1500 candidate's dataset enabling robust detection of discrimination

4.1 Technical Bias Detection

In recent years, the increasing integration of machine learning models in AI decision-making specially in recruitment context has raised the necessity of analysing fairness and potential bias (such as bias against certain groups) embedded within the data. This section aims to systematically evaluate the core analytical component of this thesis study to identify unusual patterns, measures and quantify bias by doing empirical analysis and providing evidence of where and how bias exists in the dataset, how it creates unfair or biased machine learning outcomes in the context of hiring decision.

Bias detection and fairness analysis are the crucial part of understanding the dataset and its integrity, impact, and influence on machine learning models. As historical inequalities, imbalance data representation, unaddressed discriminatory sensitive features can manifest bias in training data, ML models are using this training data to train the model and leads to unfair and discriminatory outcomes. So, exploring and identifying the sensitive features, correlation between features, dependencies between features, group-wise analysis and comparison between features will reveal whether there is any inequality or imbalance presents in the dataset. The need for this analysis is to address the core research question of this studies which is identifying critical attributes that led to biased outcomes(RQ1). It also gives the foundation of why we need to implement machine learning models to evaluate outcomes before and after implementing mitigation techniques using statistical and exploratory analysis, AI based bias mitigating tools and explainable AI techniques (RQ2, RQ3 and RQ4).

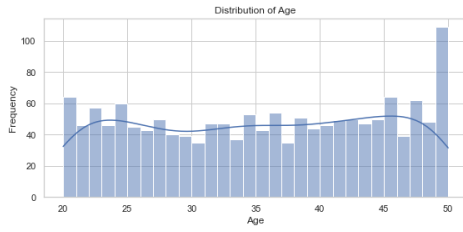
This section investigates and identifies the potential sources of biases and fairness issues within the recruitment dataset that mentioned in the earlier section of this chapter. The analysis organized into four key phases:

1. Exploratory Data Analysis based on Bias oriented recruitment data
2. Statistical Data Analysis
3. Bias Identification using Explainable XAI
4. Bias Identification using AI Tool

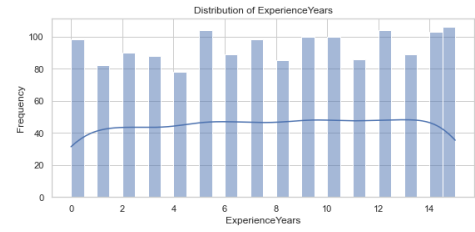
4.1.1 Exploratory Data Analysis on Bias oriented recruitment data

4.1.1.1 Uni-variate Analysis

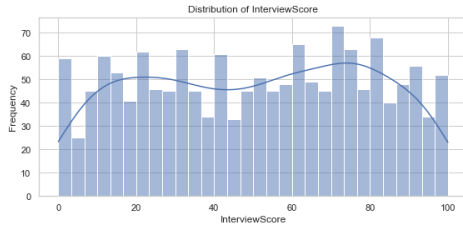
This analysis involves the distribution of individual features such as Gender, Age, ExperienceYears, InterviewScore etc, with the help of visualization in Figure-4.2. to find any skewness or irregular patterns, potential imbalance on the data using histogram and boxplot.



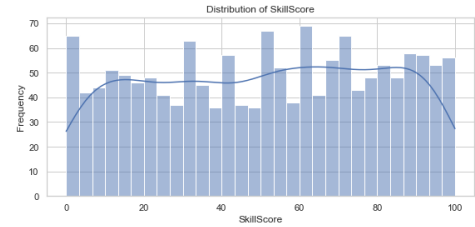
(a) Age Distribution



(b) Experience Distribution



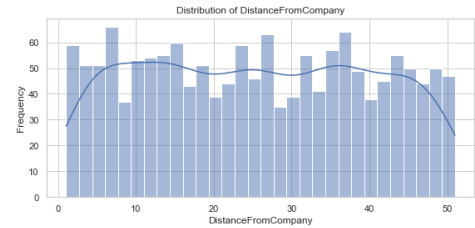
(c) InterviewScore Distribution



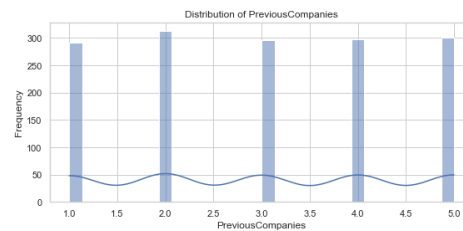
(d) SkillScore Distribution



(e) PersonalityScore Distribution

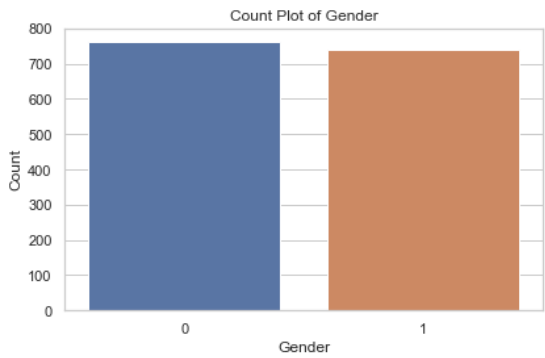


(f) DistanceFromCompany Distribution

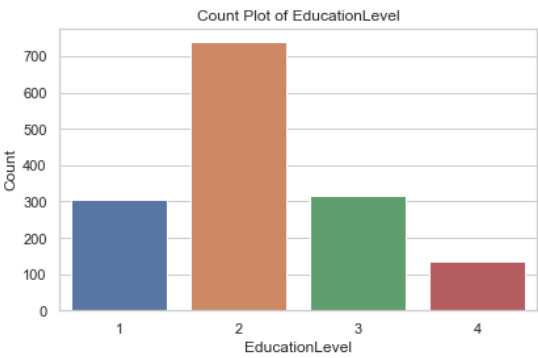


(g) PreviousCompanies Distribution

Figure 4.1: Uni-variate Distributions of Numerical Features



(a) Gender Distribution



(b) Education Level Distribution



(c) Recruitment Strategy Distribution



(d) Hiring Decision Distribution

Figure 4.2: Univariate Distributions of Categorical Features

Key findings from Uni-variate Analysis

The following Table-4.1 shows the summarize findings of uni-variate analysis of each features.

Feature	Distribution Shape	Observation
Age	Nearly Uniform	Evenly distributed across 20-45 with a spike on 50
Gender	almost balanced/ uniform	nearly equal split between male-762 and female-738 candidates
EducationLevel	Skewed	among 4 educational categories, Bachelor (Type 2) dominant strongly visible whereas fewer candidates with PhD. The distributions are Bachelor's (Type 1)-307, Bachelor's (Type 2)-746, Master's-317 and PhD-136 candidates
ExperienceYears	approximately uniform	Balanced representation from 0-15 years experience
PreviousCompanies	uniform	distributed fairly in each category(1-5 companies)
DistanceFromCompany	uniform-like	suggests company hire from both local and distant locations
InterviewScore	slightly bell-shaped	most candidates score between 70-80.
SkillScore	roughly uniform	evenly distributed suggest diverse skill background
PersonalityScore	Irregular	highest around 10-20 score, different personality types
RecruitmentStrategy	skewed	Strategy 2 dominant strongly visible. The distributions are 1: Aggressive-445, 2: Moderate-770 and 3: Conservative-285 candidates
HiringDecision	Imbalance	about 1035 candidates not hired vs 465 hired , which shows most candidates are rejected

Table 4.1: Summary of uni-variate Analysis — Numerical and Categorical Features vs. Hiring Decision

4.1.1.2 Bi-variate Analysis

The primary focus of Bi-variate analysis is to find the relationship between each feature with the target feature .especially Figure-4.3shows the leation ship of numerical features with target feature and Figure-4.4 shows relation of categorical features. This identifies whether there is any patterns, relationships, potential biases towards outcomes of the ML models by the features. This also helps to develop such a model that produce no discriminatory outcomes towards specific group.

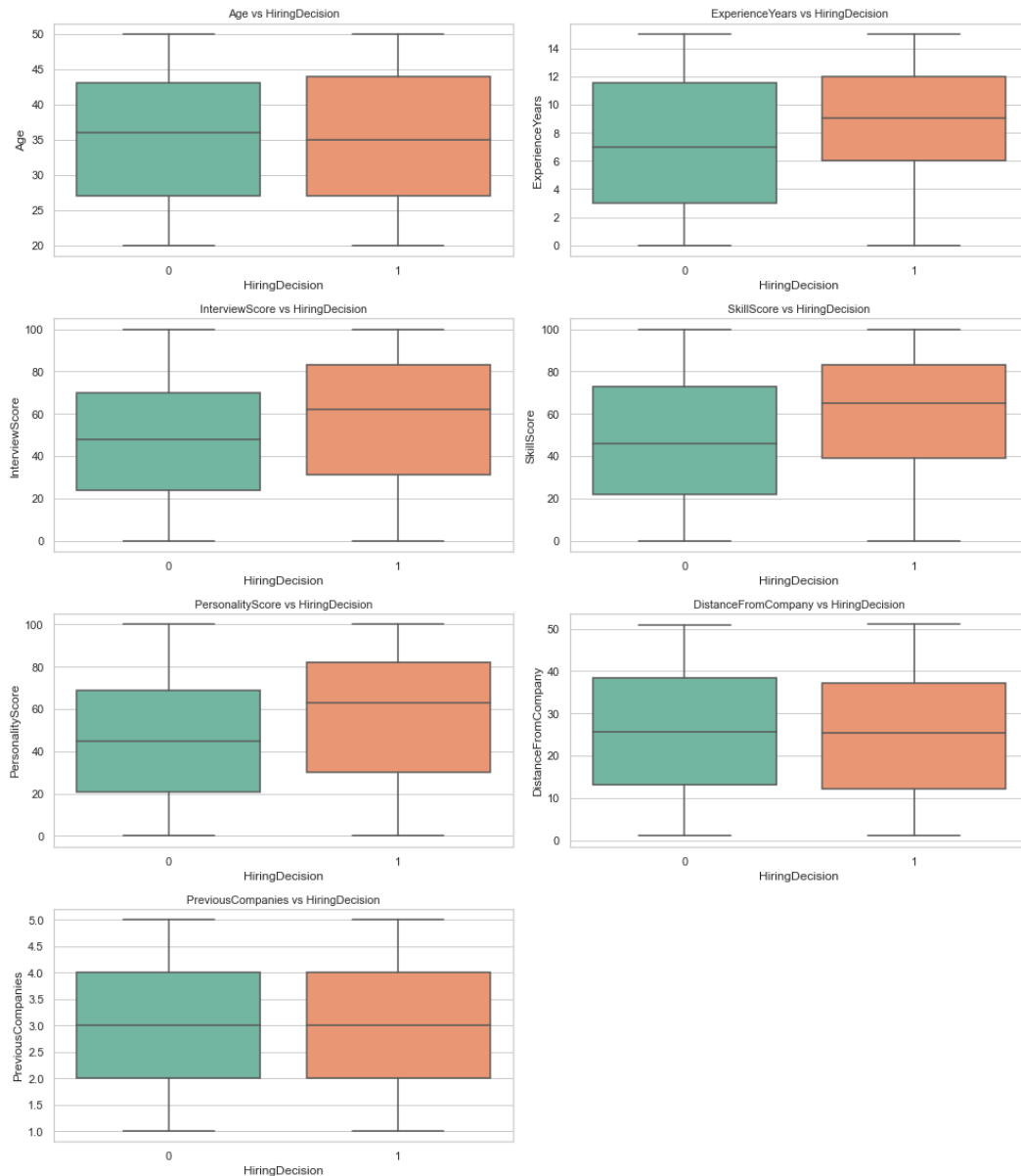


Figure 4.3: Bi-variate distribution between Numerical features vs HiringDecision

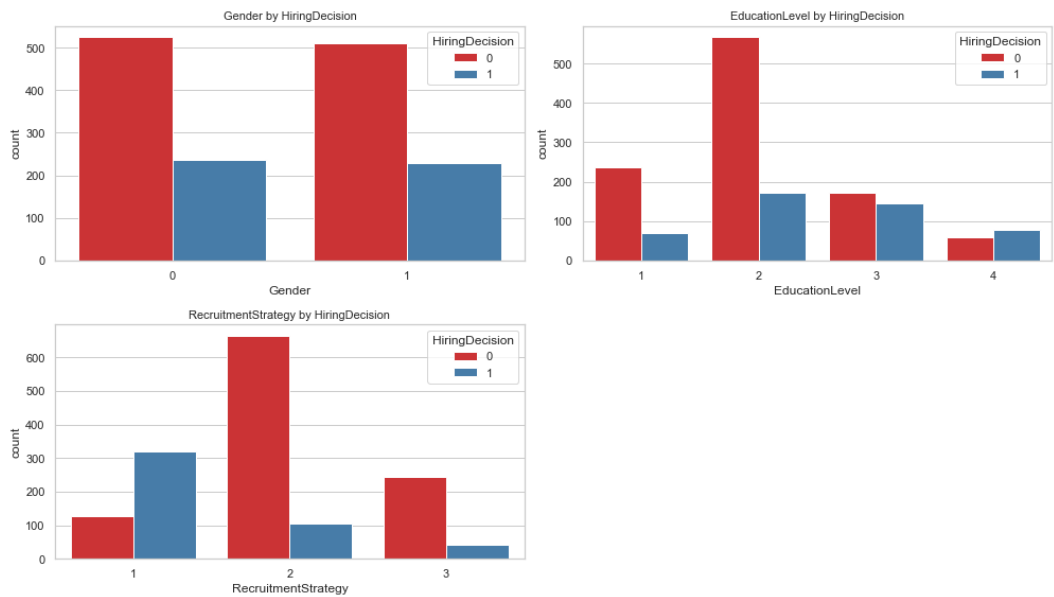


Figure 4.4: Bi-variate distribution between Categorical features vs HiringDecision

Key findings from Bi-variate Analysis

The following Table-4.2 shows the summarize findings of bi-variate analysis of each features.

Feature	Observation	Interpretation
Age	Slightly high on median among hired	Hiring may favor older (above 30) applicants.
Gender	Slight difference	No significant favor to any group
EducationLevel	Hiring rate high in higher education level (for master's and PhD candidates)	Strong relationship between ExperienceYears and HiringDecision features
ExperienceYears	Clear upward shift for hired group	Strong relationship between ExperienceYears and HiringDecision features
PreviousCompanies	No difference between groups	Past employment companies may not affect hiring
DistanceFromCompany	Slightly low on median among hired	Slightly favored to geographically closed applicant
InterviewScore	Higher median for hiring group	Strong dominant factor in hiring decision
SkillScore	Higher median for hiring group	dominant factor in hiring decision
PersonalityScore	Higher median for hiring group	dominant factor in hiring decision
RecruitmentStrategy	Strategy 1 shows higher hiring rate	Strategy 1 is more effective in hiring outcomes

Table 4.2: Summary of Bi-variate Analysis — Numerical and Categorical Features vs. Hiring Decision

4.1.1.3 Group Disparity and Fairness

This subsection aims to identify unequal and disparities outcomes by examine different demographic and categorical groups within the recruitment dataset. Features like Gender, Age, Education level and recruitment strategy are widely known as important and sensitive features so addressing these features is must to avoid legal and ethical issues related to this feature. Group disparity fairness assessment uncovers potential bias, highlights whether specific groups facing systematic advantages or disadvantages by comparing statistical measure and visual tools. Understanding group-based disparity findings is crucial because it identifies direct and indirect biases which ensures transparency and accountability in recruitment decision.

From Uni-variate and Bi-variate feature analysis, it shows that the distribution of Age, EducationLevel and RecruitmentStrategy features may affect the decision-making pro-

cess then others. So, to understand whether there are any negative effects, all these sensitive features will be compared with the key features of the dataset. The selected group attributes for disparity analysis are Gender, Education Level, RecruitmentStrategy. As Age feature is in continuous variable so to compare it in group disparity methods, it needs to be converted into categorical groups. With the help of binning, the age feature is converted into three age group categories which are:

1. Age Group 1 (20-29) consists 490 observations representing Early-career applicants.
2. Age Group 2 (30-39) consists 446 observations representing Mid-career applicants.
3. Age Group 3 (40-50) consists 564 observations representing Senior level-career applicants.

In this phase, the categorical features of the dataset including Age were analysed against hiring decision. The visual representation, shown in Figure 4.5 of the analysis indicates some unfair treatment to certain groups.

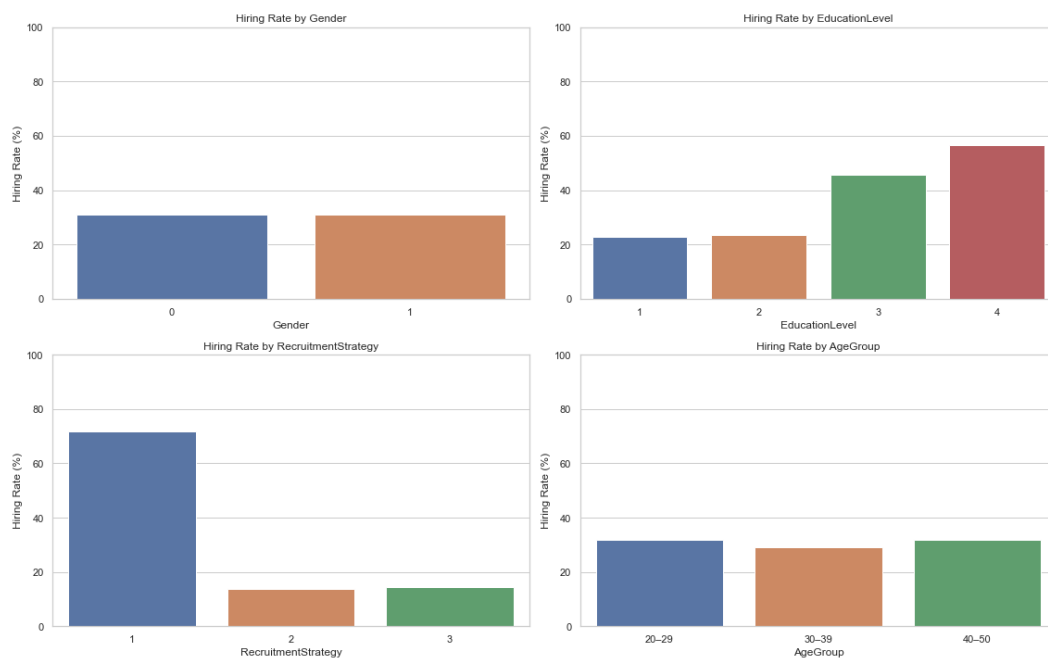


Figure 4.5: Group Disparity Analysis of Categorical Features against Hiring Decision

Disparity Analysis Summary

The Group Disparity analysis uncovers some important insights about the demographic and categorical features of the dataset. In Bi-variate analysis, almost no gender bias was observed, which is confirmed in group-wise hiring rates also (Male hiring rate 31.10%, female hiring rate 30.89%), representing consistent fairness for the feature 'Gender'. On

the other hand, from the early analysis of feature 'EducationLevel' it showed that the higher education gets priority in recruitment success which is also observed in group parity analysis as it showed category Type 4(PhD) gets 56.62% success in hiring.

The most noticable observation came from 'RecruitmentStrategy' as from Univariate analysis, it showed strategy 2 represents most of the candidates overall but in group disparity analysis it showed exceptionally high rating in Strategy 1 (Hiring rate 71.69%) which strongly suggest bias behavior towards this particular group. In Age group analysis it showed applicants aged 20-29 (hiring rate 31.84%) and 40-49 (hiring rate 31.74%) gets slightly more hiring then the group 30-39 (hiring rate 29.15%).

4.1.1.4 Correlation and Dependency Analysis

Correlation Analysis is one of the crucial step of data analysis as it reveals and examines relationships between multiple features to detect possible proxy features as these indirect features also led to indirect discrimination. This subsection also investigates dependencies between features, critical and sensitive attributes and the target feature(HiringDecision). The goal is to detect potential bias sources in the dataset by identifying variables and their direct or indirect influence on the hiring decision.

Socio-demographic features such as gender, education level etc. are the prominent source of bias in dataset as their influence through direct or proxy variables changes the model outcomes. Pearson correlation, mutual information etc techniques are generally to analyse this influence of the features.

Pearson Correlation Analysis

Pearson correlation were computed for all the variables, as shown in Figure-4.6 in the dataset to assess linear relationship and dependencies. The resulting correlation matrix visualized as a heatmap is attached in figure shows feature to feature relation correlation and dependencies with the target variable.

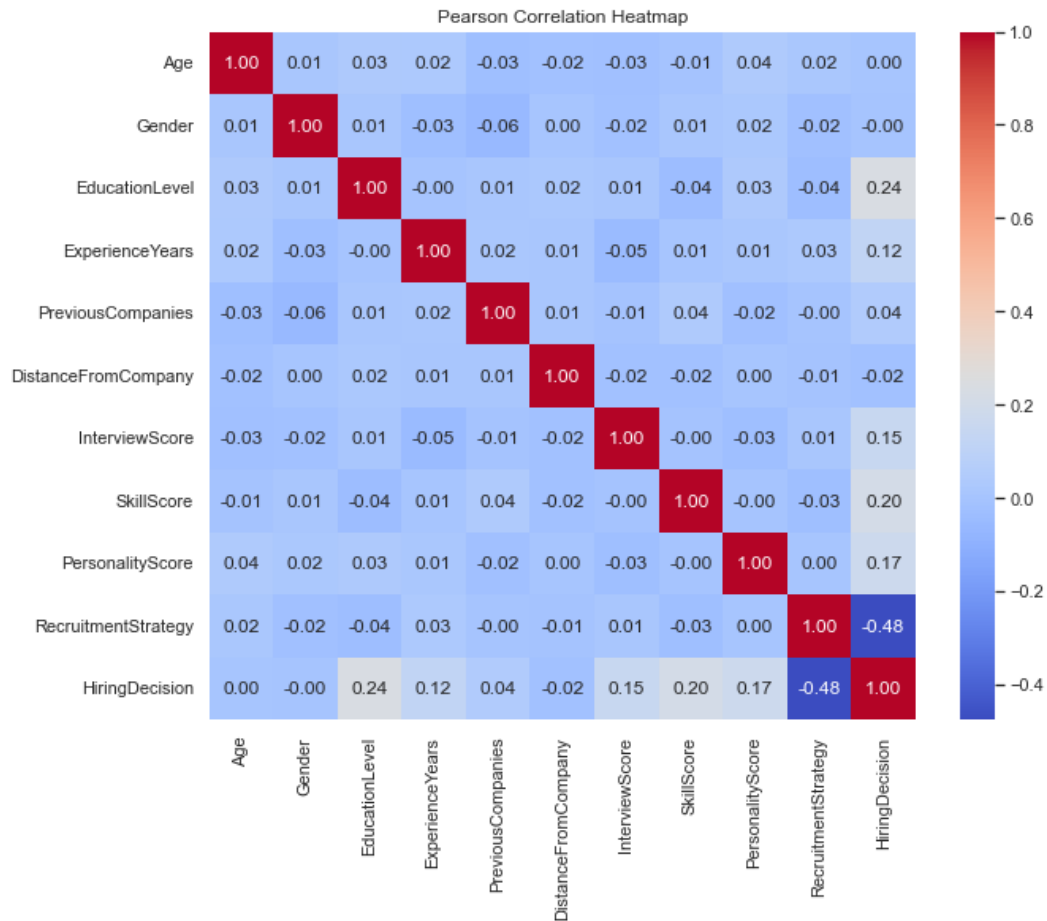


Figure 4.6: Pearson Correlation Matrix

Key observations for Continuous variables

- Most features showed low to moderate correlation.
- No multicollinearity was detected for continuous variables.
- Features like 'Age' and 'DistanceFromCompanies' showed minimum negligible correlation which illustrates that these features have no direct influence in hiring decision.

Point-Biserial Correlation

Point-Biserial correlation technique was applied with the continuous variable which is Age and the target variable 'HiringDecision' which is in binary form to evaluate the linear relationship between these features. The correlation between Age and HiringDecision feature is:

Point-Biserial Correlation (Age vs HiringDecision): $r = 0.00$, $p = 0.943$

It also indicates there is no negative correlation between age and hiring decision which eliminates the age bias concerns in hiring decision.

Mutual Information Analysis

The main goal of implementing mutual information analysis is to identify 'which features contribute how much in the hiring decision process' for both linear and non-linear features. Figure-4.7 shows the mutual information score for the numerical values within dataset.

```
Mutual Information Scores:
Age                : 0.000
ExperienceYears    : 0.032
InterviewScore     : 0.023
SkillScore        : 0.021
PersonalityScore   : 0.020
DistanceFromCompany : 0.000
PreviousCompanies  : 0.000
```

Figure 4.7: Mutual Information for Features

Key observations

- 'Age', 'PreviousCompanies' and 'DistanceFromCompany' shows almost no power in hiring outcome .
- Features such as 'InterviewScore' and 'SkillScore' also have predictive importance.

Cramér's V for Categorical Associations

Cramér's V for Categorical associations was computed as shown in Figure-4.8 to explore the relationship between categorical variables and hiring decision.

```
Gender              : 0.001
AgeGroup            : 0.026
RecruitmentStrategy : 0.571
EducationLevel      : 0.263
```

Figure 4.8: Cramér's V for Categorical Features

Key observations

- No evidence of 'Gender' bias as its acting as in independent variable .

- 'Age Group' feature has minimal influence on hiring.
- 'EducationLevel' and 'RecruitmentStrategy' showed a strong influence on the hiring decision.

Correlation and Dependency Analysis Summary

In this section, to understand and examine relationships and dependencies were examined to identify potential sources of bias. With the help of Pearson and Point Biserial correlations, it was revealed that there are some features (such as Age, DistanceFromCompany etc.) in the dataset which have minimum influence on hiring decision. Mutual Information was calculated and through MI, it was identified that 'RecruitmentStrategy', 'InterviewScore', and 'SkillScore' are the key influential predictive feature in the dataset. Cramér's V analysis also identifies a strong influence of feature like 'EducationLevel' and 'HiringDecision' which raise the issues of biasness in dataset and more analysis for mitigation strategies.

4.1.2 Statistical Data Analysis

This section evaluates hiring decisions in different demographic groups through a statistical lens using group-based fairness analysis. The goal is to evaluate whether a certain group has indifferent advantages or disadvantages as a result of potential bias exists in the dataset. Demographic Parity, Disparate Impact and Equal Opportunity metrics were implemented in this phase to identify any systematical disadvantages to any group or subgroup.

Demographic Parity

Demographic parity ensures equal probabilities of being hired regardless of the protected attributes.

Disparate Impact Ratio

Disparate Impact assesses the comparison of hiring rate between both privileged and unprivileged groups. A ratio below 0.8(80%) may indicate possible discrimination against the unprivileged or minority group.

Equal Opportunity

Equal opportunity focuses on the applicant who is truly qualified, should be hired, and ensures a true positive rate across different groups. If the true positive rate is lower, then there is possibility of bias.

Statistical Fairness Result analysis and summary

For Gender

If the difference between positive prediction and protected group is less than 0.10 or 10% , then it suggest one group is being selected significantly less, which suggest bias.

Metric	Male	Female	Observation
Hiring rate	31.10%	30.89%	Fair/ Equal
Demographic Parity		0.002	Fair/ Equal
Disparate Im-pact ratio		0.993	Fair/ Equal.
Equal Opportu-nity	1	1	Fair/ Equal

Table 4.3: Fairness analysis by Gender

Observation: There is no evidence of 'Gender' bias in hiring decision shown in Table-4.4 as all matrices are in acceptable ranges.

Education Level

From exploratory data analysis, it showed that most of the applicants selected in hiring is from category 4 (PhD). So, category 4 is used as the privileged group for comparison purposes using 80% or 0.80 rules, means that a DIR below 0.80 suggest potential bias.

Metric (category 4)	Demographic Parity	Disparate Im-pact ratio	Equal Oppor-tunity	Observation
Category 1/4	0.338	0.403	1	possible bias
Category 2/4	0.332	0.413	1	possible bias
Category 3/4	0.109	0.808	1	Fair/Equal

Table 4.4: Fairness analysis by EducationLevel

Observation: From Table-4.4, it shows possible bias towards lower educational level means hiring preference in higher studies, which suggests education bias in dataset.

For Recruitment Strategy

From exploratory data analysis, it was shown that Strategy 1 is the most effective recruitment medium, as most of the applicants were selected from this source. So, Strategy 1 is used as privileged group for comparison purposes using ≤ 0.10 rules, a mean difference greater than 0.10 suggest potential bias.

Metric (category 1)	Demographic Parity	Disparate Impact ratio	Equal Opportunity	Observation
Category 2/1	0.580	0.190	1	possible bias
Category 3/1	0.573	0.201	1	possible bias

Table 4.5: Fairness analysis by EducationLevel

Observation: Table-4.5, strongly suggests possible bias towards Strategy 2 and 3 as they have significantly lower hiring rates.

4.1.3 Bias Detection Using Explainable AI

4.1.3.1 Bias detection insight from Exploratory and Statistical analysis

To understand and identify the source of bias in dataset, a series of exploratory and statistical data analysis techniques were implemented to the dataset in Chapter 2-Methodology. These analyses helped to identify disparities within the dataset specially uncovers differences treatment between categorical groups. The analysis results informed and urged to more in-depth granular interpretation using Explainable AI (XAI) for better understanding of the features relationship with hiring decision within the dataset. The key finding from EDA and statistical analysis are:

- Uni-variate analysis showed that the dataset contains more observation for Class 2(Bachelors (Type 2)) of EducationLevel feature and Strategy 1(Moderate) RecruitmentStrategy.
- Across all model, Class 0 (which is ‘ Not Hired’) constantly showed higher precision, recall and F1-score.
- However, in Group-based disparities showed that despite having small number of observations of class 4(PhD), the candidates received disproportionately higher hiring decisions, which makes it as a privileged group of that feature. Additionally group-based disparity also suggests that though Strategy 2 (Moderate hiring strategy) had more observation in RecruitmentStrategy feature but the most positive hiring came from Strategy 1(Aggressive hiring strategy).
- From correlation analysis, it was revealed that Age, Gender, PreviousCompanies features has no strong relation with hiring decision.
- In addition, from mutual Information analysis, it revealed that the assessment score of a candidate also has influential relationship with hiring decision
- Finally Cramér’s V analysis, showed that the most influential variable linked with hiring decision are EducationLevel and RecruitmentStrategy.

From these findings, it was clearly revealed that the usual traditional protective features such as Age, Gender might not have strong influence in hiring decision whereas the non-traditional features like EducationLevel, RecruitmentStrategy have potential reasoning being a biased feature. However, only statistical and group level findings don't validate the source of bias within a dataset as these techniques can't reveal the reasoning of how a model predict one outcome. So, to address this limitation, we implemented Explainable AI techniques to understand 'instance-level' and 'global-level' interpretations of ML model predictions

4.1.3.2 Implementation of Explainable AI to Identify Bias

This section aims to explain the decision-making process of Random Forest and Decision Tree classifier using LIME and SHAP. Among the five classifier Decision Tree and Random Forest were selected for further in-depth interpretability for various reasons:

- **Strong Baseline Performance:** Both models predicted strong performance for the majority class which is predicting 'Not Hired' class.
- **Compatibility with XAI:** LIME and SHAP technique will be compatible with the inherent structure of rules-based decision process.
- **Transparency and Robustness:** Decision-tree provides traceable path to each prediction, make the model more transparent which is crucial to domain like recruitment for ensuring accountability. At the same time, as Random Forest combines decision of multiple trees, it increase the robustness of the model. Both transparency and robustness offer deeper understanding of the bias source and decision making.

4.1.3.3 Implementation of Explainable AI for Random Forest

LIME Analysis

Global Feature Importance

To determine which features, have most influence on the hiring decisions, this study analysed the average LIME contribution for both decision tree and random forest of each feature s. The goal is to find which variables shape the model outputs significantly.

Top 10 Most Influential Features:

1. RecruitmentStrategy	Importance: 0.3944	Variability: 0.4129
2. EducationLevel	Importance: 0.1838	Variability: 0.0535
3. InterviewScore	Importance: 0.1214	Variability: 0.4411
4. SkillScore	Importance: 0.1168	Variability: 0.0693
5. ExperienceYears	Importance: 0.1040	Variability: 0.2762
6. PersonalityScore	Importance: 0.0973	Variability: 0.2396
7. PreviousCompanies	Importance: 0.0206	Variability: 0.3617
8. DistanceFromCompany	Importance: 0.0091	Variability: 0.7693
9. Age	Importance: 0.0087	Variability: 0.6573
10. Gender	Importance: 0.0058	Variability: 0.7390

Figure 4.9: Feature Importance using LIME

As shown in Figure-4.9 , the most influential features were RecruitmentStrategy, EducationLevel,InterviewScore and SkillScore consistent with Mutual Information analysis. However, it also raises the question of RecruitmentStrategy and EducationLevel features may disproportionately influence decision outcomes as these has highest importance.

Group-level Disparity for EducationLevel

As multiple analysis showed EducationLevel as a potential source of bias, implementing group level disparity examine the contribution of this feature based on education level as shown in Figure-4.10.

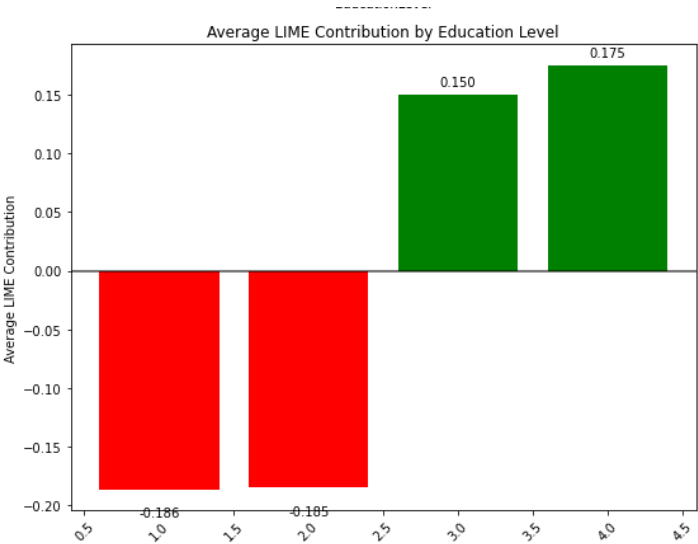


Figure 4.10: LIME contribution by Education Level


```

Most Privileged Education Level: 4.0 (Avg LIME: 0.175)
Least Privileged Education Level: 1.0 (Avg LIME: -0.186)

=== EFFECT SIZE (4.0 vs 1.0) ===
Cohen's d for LIME contributions: 46.249
Large effect size - Strong bias detected

```

Figure 4.11: Bias in Education Level using LIME

Figure-4.11 shows that level 1 (bachelor's 1) are least privileged candidates, whereas as level 4 (PhD) candidates are the privileged one. This also proves the EDA analysis of level 4 (PhD) class being the Privileged group.

Group-level Disparity for RecruitmentStrategy

For recruitment strategy, LIME techniques also revealed that strategy 1(Aggressive) which is clearly shown in Figure4.12 gets more hiring predictions compare to the other classes.

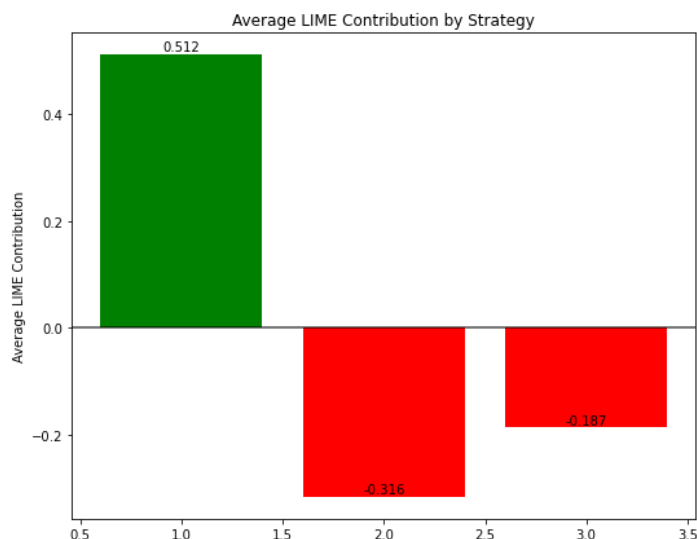


Figure 4.12: LIME contribution by Education Level

```

Most Privileged Strategy: 1.0 (Avg LIME: 0.512)
Least Privileged Strategy: 2.0 (Avg LIME: -0.316)

=== EFFECT SIZE (1.0 vs 2.0) ===
Cohen's d for LIME contributions: 118.845
Large effect size - Strong bias detected

```

Figure 4.13: Bias in Recruitment Strategy using LIME

These patterns validate the initial fairness analysis and also suggest that there may exist systematic model preference towards recruitment strategy 1(Aggressive).Figure-

4.13 also validates the initial analysis where it recruitment strategy 1 was identified as domination class in hiring decision process.

Local Explanation

A local lime explanation for rejected candidates are shown below figure to prove the education level biasness.



Figure 4.14: Hiring Decision for a Candidate

From Figure- 4.14, it shows that the candidate has low performance scores (personality, skill, interview), yet the model still predicts a positive outcome ‘hired’ with 100% prediction probabilities. It shows that the model heavily biased towards Education level and also gets influence by previous experience years of a candidate.

Intersectional Bias Analysis via Clustered LIME

To investigate whether there is any hidden pattern available in the dataset, an intersectional analysis shown in Figure- 4.15between the top two influential features was conducted. The main goal of this clustering analysis is to find the interaction between privileged groups, not just isolated features.

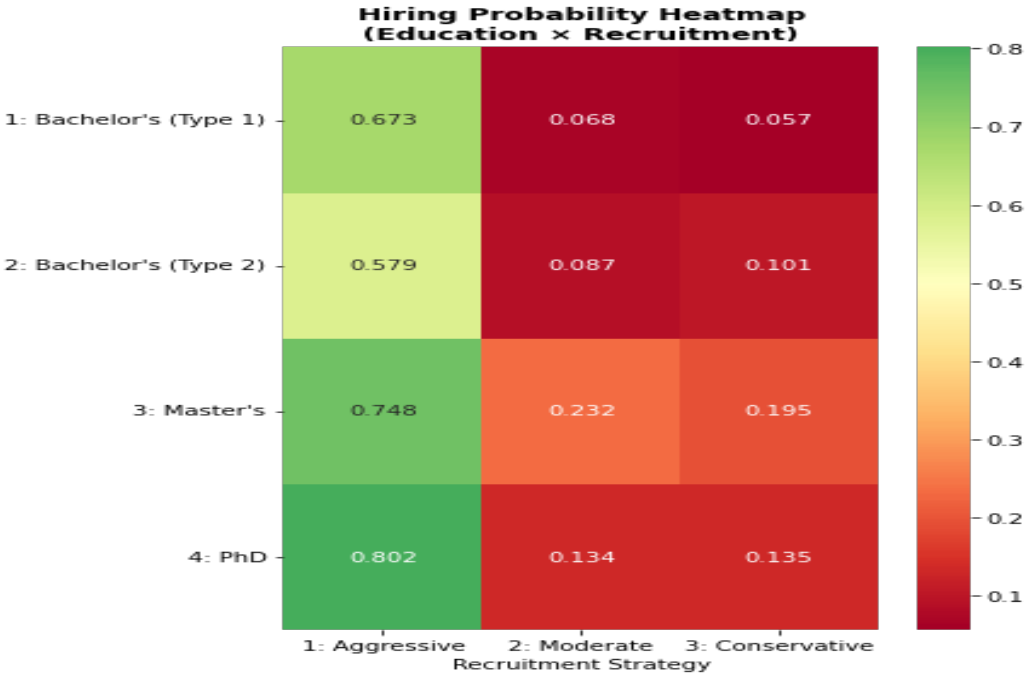


Figure 4.15: Heatmap of Intersectional Cluster

The analysis showed that a candidate with a PhD (level 4) and aggressive (strategy 1) recruitment category always received highest average hiring rate. That means there is 80% probability of this cluster to get hired which shows a strong model preference.

SHAP Analysis

Global SHAP Summary

The SHAP value top features influencing hiring decision also confirms the influence hierarchy seen in the LIME analysis in Figure-4.16.

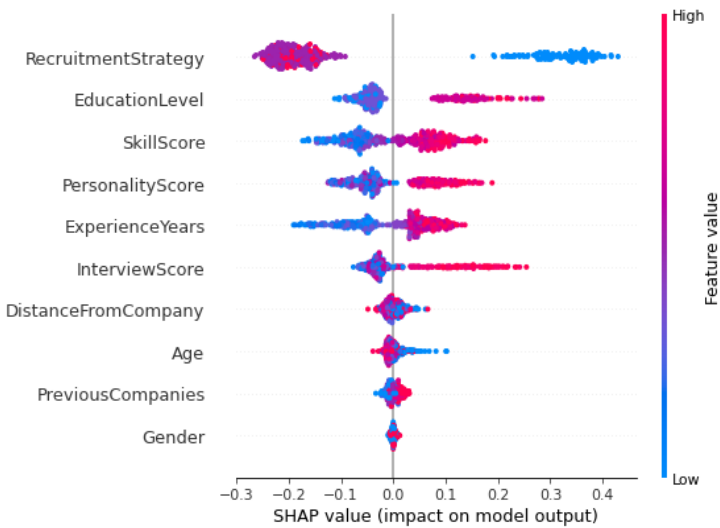


Figure 4.16: Global SHAP impact summary

The top most impactful and influential features were Recruitment Strategy, Ed-

ucation Level, Skill Score, Personality Score, and Experience Years.. The consistent influential presence of education level and recruitment strategy among top features behind decision highlights their role in model bias.

SHAP Feature Importance

The SHAP top features ranking illustrates in Figure-4.17the average contribution of each feature.

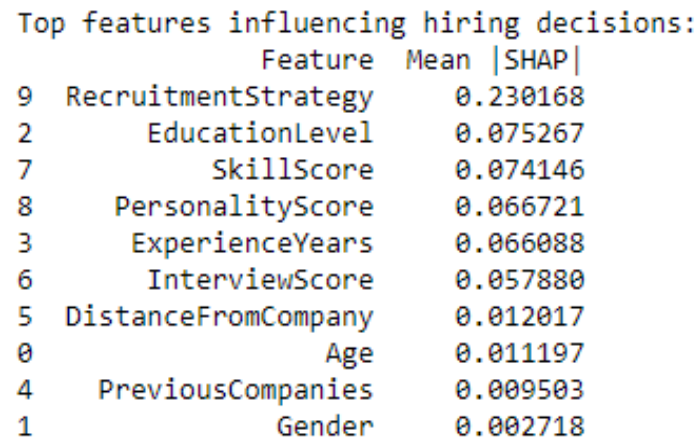


Figure 4.17: SHAP Top features

However the traditional bias-prone feature like Gender and Age appear at the bot- tom of the list, with minimal SHAP values, supporting both LIME and earlier fairness metrics (EDA and statistical analysis) indicating minimal gender and age bias.

SHAP Force Plot Local Interpretation

The SHAP force plot as shown in Figure-4.18 for a specific prediction where the can- didate was not hired.



Figure 4.18: SHAP Force plot for candidate hiring decision

Although the candidate had high Skill Score and 15 years of Experience Years, but the model output was lowered by the candidate’s Recruitment Strategy 2 (Moder- ate) and Education Level 2 (Bachelor). This plot confirms how group-level disparities (clustered group) identified earlier can directly affect individual outcomes.

4.1.3.4 Explainable AI for hiring decision using Decision Tree

Random candidate local explanations In this section, The Table-4.6 shows three different candidates hiring decision, using LIME (in decision tree) to understand the depth of the influence of certain features, such as the recruitment strategy over hiring decision.

Candidate Information	Actual Outcome	Model Prediction	Comments
Gender: Male Age: 48 Education Level: 2.0 Experience: 15 yrs Previous Companies: 1 Distance: 29.6 km Interview Score: 40/100 Skill Score: 69/100 Personality Score: 53/100 Recruitment Strategy: 2.0	Not Hired	Not Hired	Though the candidate had 15 years of experience with a skill score of 69, due to recruitment strategy and education level, the model predicted 'Not Hired'.
Gender: Male Age: 34 Education Level: 2.0 Experience: 10 yrs Previous Companies: 1 Distance: 7.2 km Interview Score: 33/100 Skill Score: 20/100 Personality Score: 82/100 Recruitment Strategy: 1.0	Hired	Hired	Despite a low skill score of 20 and interview score of 33, the candidate was hired, possibly due to being part of the privileged recruitment strategy group.
Gender: Female Age: 48 Education Level: 2.0 Experience: 9 yrs Previous Companies: 3 Distance: 30.1 km Interview Score: 36/100 Skill Score: 91/100 Personality Score: 25/100 Recruitment Strategy: 2.0	Not Hired	Not Hired	Despite having a high skill score and reasonable experience, the candidate was rejected likely due to low interview and personality scores and unprivileged recruitment strategy.

Table 4.6: Examples of individual predictions and model reasoning based on candidate features

4.1.4 Bias Detection Using AI Tool

To detect possible potential bias in recruitment dataset, two popular widely used AI fairness toolkits -AIF360[aif360] and Fairlearn[48] will be implemented in this phase to evaluate group based disparities across protected features. The analysis was conducted in data level, without any prior machine learning modeling to understand the actual structural pattern or imbalances present in the dataset which could influence the outcome of hiring decision. The analysis examined 6 different features from the dataset as from previous data analysis method showed that few features within the dataset may introduce bias. So based on the previous finding and standard socially sensitive features criteria these features were selected for further analysis. The attributes are Gender, EducationLevel, Age(using age groups), ExperienceLevel, DistanceFromCompany, and RecruitmentStrategy.

Dataset Configuration

For AIF360

Before implementing AIF360, the dataset was converted in AIF360 format, which is 'BinaryLabelDataset'. This format helps to identify the target variable and the protected or sensitive features of the dataset. Throughout the implementation, fairness metrics were applied consistently while preserving the key words such as privileged and unprivileged group. For thorough bias assessment using AIF360, the complete recruitment dataset including 1500 candidate observation and 11 features were used.

For Fairlearn

Unlike AIF360, Fairlearn doesn't required any additional wrappers for bias detection. Instead, Fairlearn works directly with standard Python object such as 'NumPy' arrays and 'pandas' series as a result it provides faster and rapid implementation of dataset data-level fairness analysis.

Feature Engineering for AIF360 and Fairlearn

As part of the data preparation to implement group-based fairness analysis of both AI toolkit, certain continuous variables were transformed into categorical groups specially in binary form: 1(privileged group) and 0(unprivileged group), as shown in Table-4.7. For instance,

Feature	Privileged Group	Unprivileged Group
Age	≥ 35 years (Older)	< 35 years (Younger)
Education Level	Levels 3 and 4 (Higher education)	Levels 1 and 2 (Lower education)
Recruitment Strategy	Strategy 1 (Aggressive)	Strategies 2 and 3 (Moderate, Conservative)
Distance from Company	≤ 25.5 km (Local)	> 25.5 km (Remote)
Experience Years	≥ 8 years (Senior)	< 8 years (Junior)

Table 4.7: Privileged and Unprivileged Group Definitions for Fairness Analysis

Bias Detection and Analysis for each feature

Gender (Protected feature) Bias Assessment

The dataset contains 762 male candidates (50.8%) and 738 female candidates (49.2%). The following Table-4.8 summarizes fairness evaluation results using AIF360 and Fairlearn for the feature 'Gender'.

AIF360 result	Fairlearn result
Statistical Parity Difference: 0.0021	Demographic Parity Difference: 0.0021
Disparate Impact: 1.0067	Demographic Parity Ratio: 0.9933
No significant Bias detected	

Table 4.8: Bias Detection Results for Gender Feature

Assessment: The hiring rate among male candidates was 31.1% (237 out of 762), while the rate for female candidates was 30.9% (228 out of 738). The recruitment process treats both gender candidate equally at the dataset level.

Age (Protected feature) Bias Assessment

During the Exploratory Data Analysis, age was categorized into 3 categories, which are 20–29 (early career), 30–39 (mid career), and 40–50 (senior level) to identify potential bias. Though there is minor difference presents in different categorize but nothing substantial enough to suggest age bias.

So for AIF360 and Fairlearn toolkit, age will be again categorized, but this time it will be in binary group, younger candidates (under 35 years) and older candidates (35 years and above) as mentioned above. The following Table-4.9 summarizes fairness evaluation results using AIF360 and Fairlearn for the feature 'Age'.

AIF360 result	Fairlearn result
Statistical Parity Difference: 0.0086	Demographic Parity Difference: 0.0086
Disparate Impact: 1.0281	Demographic Parity Ratio: 0.9727
No significant Bias detected	

Table 4.9: Bias Detection Results for Age Feature

Assessment

The hiring rate in younger candidates are 30.6%, where as in older candidates it's 31.6%. The recruitment process treats candidates of all age fairly and equally at the dataset level.

Education Level Bias Assessment

As AIF360 evaluates the feature in binary form , so for the purpose of the assessment of bias, 4 education categories will be divided into 2 groups. Based on previous analysis, it showed that most of the hiring came from candidates with master's (category 3) or PhD (category 4) degree. As a result, both the master's and Phd education levels will be considered as privileged group, and both the bachelor (category 1 :type 1) and bachelor (category 2: type 2) will be considered as unprivileged group.

AIF360 result	Fairlearn result
Statistical Parity Difference: -0.2580	Demographic Parity Difference: 0.2580
Disparate Impact: 0.4736	Demographic Parity Ratio: 0.4736
Significant Bias detected	

Table 4.10: Bias Detection Results for EducationLevel Feature

Assessment

From fairness evaluation Table-4.10, it shows education level bias clearly. The hiring rate for higher education level is 49.01% (Groups 3,4: 222/453 candidates), where as in for lower education level it's 23.21% (Groups 1,2: 243/1,047 candidates). To understand more precisely, detailed education level hiring rate was calculated to understand which level gets maximum benefits, or influence the decision.

EducationLevel	Total candi- dates	Hired candi- dates	Hiring Rate
1	307	70	0.228
2	740	173	0.234
3	317	145	0.457
4	136	77	0.566

Table 4.11: Detailed Bias Assessment in EducationLevel using AIF360

Detailed educational level breakdown, shown in Table-4.11 detected critical bias present in this feature which created discrimination for lower education candidates, as a result they experience systematic disadvantages.

Experience Level Bias Assessment

The distribution of candidates experience shows, out of 1500 candidates, 727 had less than 8 years of experience (< 8 years), while 773 had 8 or more years of experience (≥ 8 years).

AIF360 result	Fairlearn result
Statistical Parity Difference: -0.1078	Demographic Parity Difference: 0.1078
Disparate Impact: 0.7025	Demographic Parity Ratio:0.7025
Moderate ExperienceLevel Bias detected	

Table 4.12: Bias Detection Results for ExperienceYear Feature

Assessment

From fairness evaluation Table-4.12, it shows moderate level education bias clearly. The junior or less experienced candidates hiring rate is 25.45%, while more experienced candidates hiring rate is 36.22. Although some job recruitment explicitly require prior experience for the certain position purpose. On the other hand, constantly favoring high experience candidates introduce systematic disadvantages for early-career applicants which creates experience based discrimination.

Geographic Location (DistanceFromCompany) Bias Assessment

After grouping Distance from company feature in binary form, it showed out of 1500 candidates, 1262 live more than 25.5 km from the company (> 25.5 km) known as remote group, while 238 candidates live within 25.5 km (≤ 25.5 km) known as local group.

AIF360 result	Fairlearn result
Statistical Parity Difference: -0.0040	Demographic Parity Difference: 0.0040
Disparate Impact:0.9872	Demographic Parity Ratio: 0.9872
No significant Bias detected	

Table 4.13: Bias Detection Results for DistanceFromCompany Feature

Assessment

From the result of fairness evaluation Table-4.13, it shows no bias. Geographic location doesn't create discriminatory behavior doesn't influence the hiring decision. As, The hiring rate for the local candidates, ≤ 25.5 km was 31.2%, while the rate for the remote candidates, > 25.5 km) was 30.8%.

Recruitment Strategy Bias Assessment

For the purpose of the assessment of bias, strategy 2: Moderate and 3: Conservative will be merged (1055/ 1500 candidates after merging) based on previous analysis. During exploratory data analysis, it showed strategy 1: Aggressive strategy (445/1500 candidates) influenced the hiring decision maximum.

AIF360 result	Fairlearn result
Statistical Parity Difference: -0.5785	Demographic Parity Difference: 0.5785
Disparate Impact:0.1930	Demographic Parity Ratio: 0.1930
Severe Bias detected	

Table 4.14: Bias Detection Results for RecruitmentStrategy Feature

Assessment

From the fairness evaluation presents in Table-4.14, Recruitment strategy bias represents the most extreme bias. To understand more precisely, detailed recruitment strategy shown in Table-4.15 hiring rate was calculated and found strategy 1 hires 71.69% candidates, introduces systematic disadvantages

RecruitmentStrategy	Total candi- dates	Hired candi- dates	Hiring Rate
1	445	319	0.72
2	770	105	0.14
3	285	41	0.14

Table 4.15: Detailed Bias Assessment in RecruitmentStrategy using AIF360

Key Finding and Bias severity Classification

Feature	AIF360 Bias Analysis	Fairlearn Bias Analysis	Key Finding
Gender	No Bias	No Bias	Gender groups have nearly equal hiring rates (31.1% vs. 30.9%)
Age Group	No Bias	No Bias	Minor difference in hiring rate between younger and older groups(30.6% vs 31.6%)
Education Level	Significant Bias	Significant Bias	Higher education group (Levels 3/4) hired at double the rate of lower education group (49.01% vs. 23.21%)
Experience Years	Moderate Bias	Moderate Bias	Senior candidates (≥ 8 years) had noticeably higher hiring rates than juniors (36.22% vs. 25.45%)
Geographic Distance	No Bias	No Bias	No strong hiring difference between local and remote candidates (31.2% vs. 30.8%)
Recruitment Strategy	Severe bias	Severe Bias	Aggressive strategy (1) dominates hiring, others groups are underrepresented (71.69% vs. 13.84%)

Table 4.16: Bias Severity Classification and Key Findings from AIF360 and Fairlearn

4.1.5 Summary of Bias Findings from EDA and Statistical Analysis

This subsection summarizes the key finding from the study of exploratory data analysis, statistical analysis and targeted analytical questions with the goal of identifying the source or features that may cause bias outcome.

Gender:

Hiring rates for gender feature are well distributed, with minimal parity difference in fairness metrics which suggest no gender bias in dataset.

Age and DistanceFromCompany:

Both features shows no significant linear correlation with hiring decisions.

Education level:

A strong influence and association was observed between education level and hiring decision. Category 4(Phd) candidates had significant higher hiring rate then the others. Both demographic parity and disparate impact metrics shows that candidates with education level categories 1 and 2 were the unprivileged group in 'EducationLevel' feature, where candidates of category 3(Masters) also get satisfactory hiring rate compared to the privileged category 4 group.

ExperienceYears and PreviousCompanies:

Candidates with more experience (equal or more than 8 years) were more favored, and moderate bias was observed.

Number of previous company's feature was also analysed to reveal if there is any relationship between switching jobs within experience years in hiring decisions. However, there was no significant difference in hiring was observed based on the numbers of companies worked, in fact diverse experience (3 to 5 companies worked) was also present in the successful hiring.

Interview, Skill and personality Score:

Interview and skill score assessment features are essential indicator for a good candidate in hiring decisions. Multiple analysis show that high interview score constantly aligned with positive hiring decision. During one of the analytical questions analyses, it shows that there are candidates who have low education level, but with high score of interview and skill, the education level was compensate.

Recruitment Strategy:

Hiring rate significantly dependent on this feature specially strategy 1 had more suc-

successful hiring than the others which indicates structural bias within the dataset.

Though the dataset doesn't contain direct age or gender bias, but it has indirect bias which raises the need to implement the fairness-aware models to make the dataset more transparent and fairer towards all of its groups.

4.1.6 Deep-Dive Analytical Exploration of Key Recruitment Questions

This section presents some targeted core questions that arise in recruitment context, will be investigated those questions based on the relationship between the variables in the dataset. These questions are designed using exploratory data analysis (EDA) and statistical analysis outcomes. Through data-driven question and visualization, the goal of this section is to understand not only the bias behaviour but also how the attributes (especially critical attribute) interact with one another. The questions are:

Q1: How do interview scores differ between age groups?

Goal: The goal is to identify whether some age groups are privileged or underprivileged and receive a higher or lower interview score.

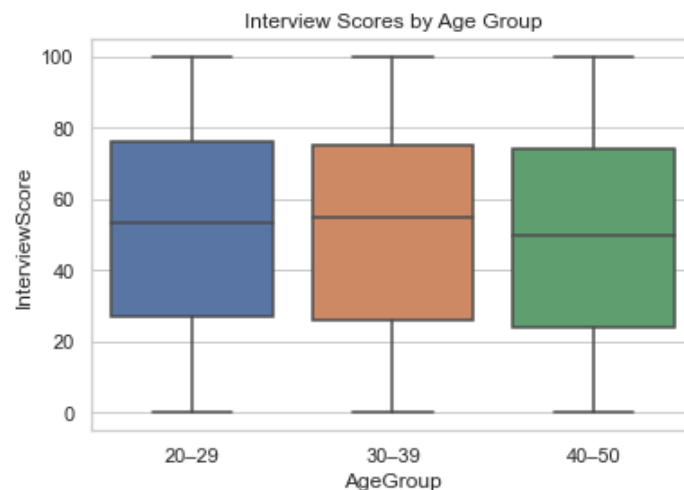


Figure 4.19: Interview scores by Age group

Evaluation: From the visualization, it showed that no group gets advantages or disadvantages in the interview because of their age.

Q2: Are there gender disparities in the interview?

Goal: To check if there is any disparity in the interview score because he is male or female.

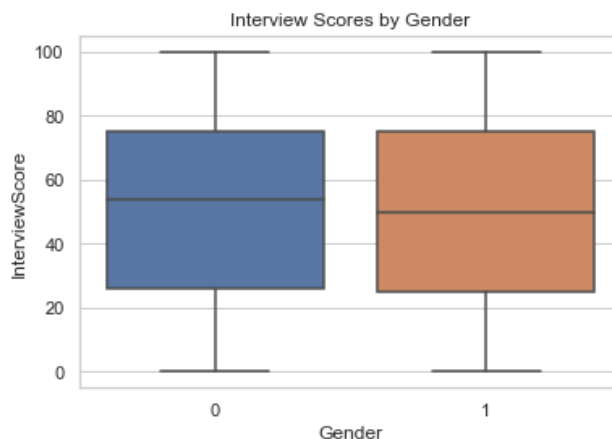


Figure 4.20: Interview scores by Gender

Evaluation: There is no major difference in values, minor differences found only so it appears interview score as gender-neutral.

Q3: Does academic qualification compensate experience or skill scores in hiring outcomes?

Goal: The goal of analysing this question is to find whether higher education(Phd) can compensate lower technical skill.

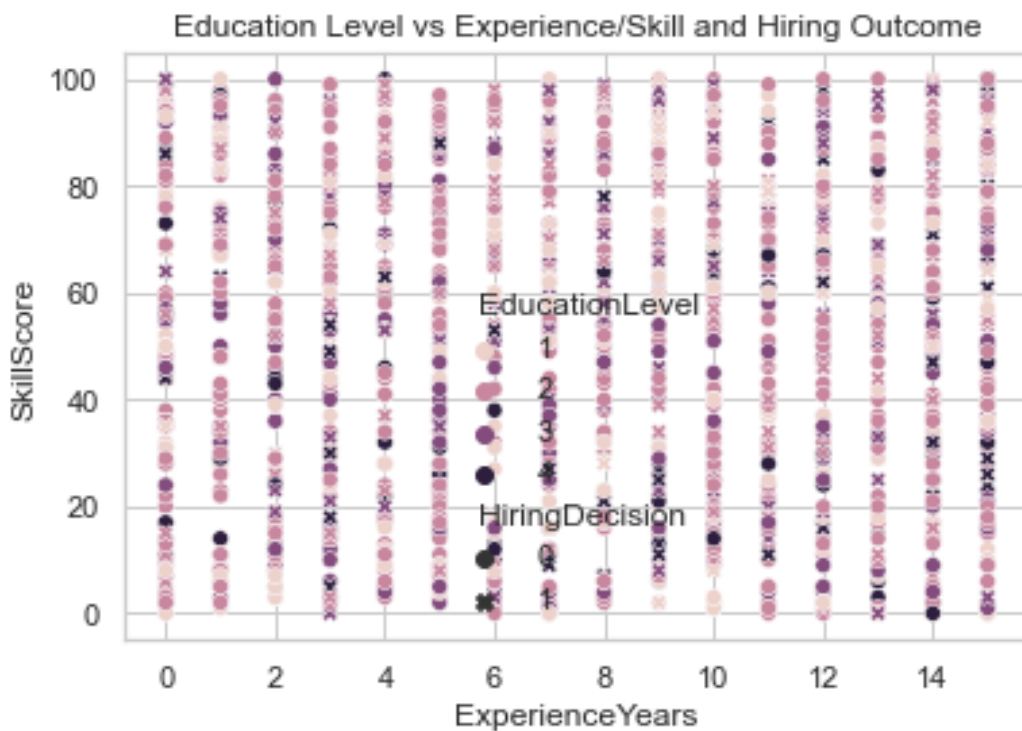


Figure 4.21: EducationLevel Vs Experience and skill in hiring

Evaluation: From the plot, it suggest that skill score plays an important role in

hiring. It shows if the skill score is high and Education level is low , yet the candidate hiring outcome is positive. So this proves with strong technical skill, the education qualification can be compensate.

Q4: Does stable job histories preferred in hiring?

Goal: One of the interesting analysis questions is to examine whether switching jobs can affect hiring results. If so, it will be a proxy variable combining experience years and number of companies worked before.

CompanyBand	1-2	3-5
ExpBand		
0-5	21.938776	20.731707
6-10	30.726257	43.686007
11-15	35.294118	33.887043

Figure 4.22

Evaluation: After the evaluation, it shows that common belief of 'job stability matters' hypothesis proven wrong in the recruitment dataset. In fact candidates with experience year (5-10) with (3-5) previous companies had the highest hiring rate. It also suggest that working in different companies may indicate diverse working experience valued in the hiring decision.

Q5: Does geographic closeness to the company have higher hiring chances?

Goal: The goal is to identify whether geographic proximity influences hiring decision such as candidates who live closer to the company get more hired or not.

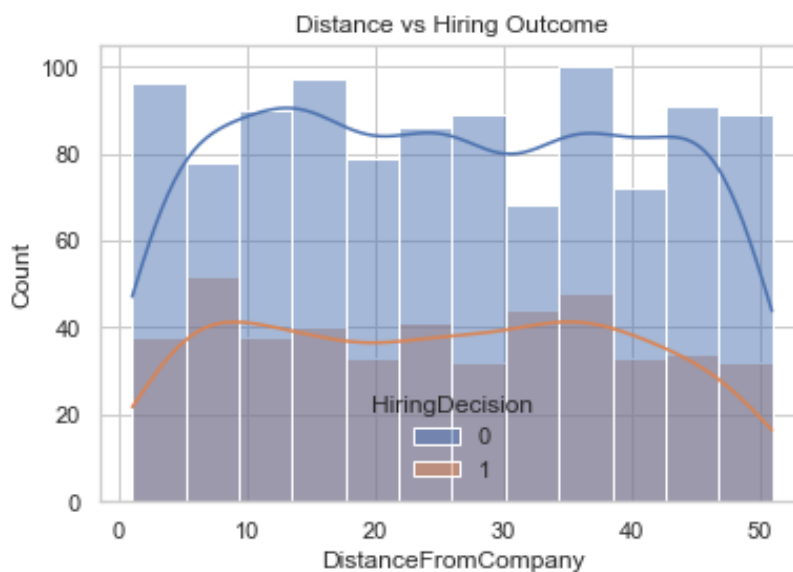


Figure 4.23: Distance Vs Hiring outcomes

Evaluation: From the Histogram plot , it shows there is no hiring bias based on geographic location.

Q6: Do recruitment strategy favored in hiring over interview score?

Goal: The goal is to find whether specific recruitment strategy always get high hiring rate, even when the candidate interview score is low. This question specifically targets both of the prominent and influential features in the dataset.



Figure 4.24: Interview score Vs recruitment strategy in hiring

Evaluation: The box plot shows hiring decision is well connected with the interview score feature. From all strategies, applicant sourced from strategy 1 yielded highest hiring rates and from the visualization it shows strategy 1 also associated with constant high interview performance.

4.2 Research Question 01

To address the first research question ‘RQ1-What are the most critical features that contribute Bias in Dataset?’- a multi level data-driven analysis was conducted. The aim of each approach such as exploratory data analysis, and statistical analysis were to uncover and evaluate all the attributes to identify the source of bias within the dataset. The key findings are:

Class distributional Imbalance:

The target variable presents class imbalance, where the ‘Not Hired’ candidates’ observation was higher then the ‘Hired’ candidates which may cause skewing model learning.

Bias-prone influential attributes:

Education level, Recruitment strategy, Experience years ,Interview score and Skill

score features are influential features then the other feature as a result hiring is strongly dependent on these features. However, candidates assessment score are more objective indicator for the recruitment purpose as individually the assessment score didn't show any strong negative influence like others.

Finally, the attributes most associates with hiring decision and shows bias towards certain group or candidate hiring decision within dataset are-

- RecruitmentStrategy
- EducationalLevel
- ExperienceYear
- And class imbalance of target attribute, which gives a insight that always bias doesn't occur from the sensitive features like Age and Gender. These findings state the importance of mitigation strategies which will be discussed in later sections.

4.3 Non-Technical approach

The non-technical approach is more of a prevention technique that is done before diving into the technical aspects, like dataset preprocessing, model building, etc. So the non-technical approach includes a bias-aware data collection, which can make the dataset bias-free from the beginning. The growing reliance on artificial intelligence in this age of advanced technology presents a number of potential hazards. One of the main issues with artificial intelligence is its bias. A biased dataset may be the first sign of bias. Bias-aware data collecting is one way to prevent bias in datasets. In order to guarantee representative and equitable datasets, bias-aware data collection is a methodical approach to data collection that proactively detects, tracks, and reduces various types of bias during the collection process. The reason why we need to address non-technical bias mitigation is the incorporation of ethical design practices, responsible data collection, policy enforcement and used education and communication. In 2016, Microsoft launched Tay, an AI chatbot to interact with twitter user, within 24 hours it started to post racist tweets as it learned from other users interaction[43]. The reason behind this failure of AI chatbot is there was no non-technical safeguards and Microsoft didn't addressed how chatbot Toy should respond to sensitive content. This case showed that bias mitigation must go beyond addressing the presence of bias in dataset or in algorithms and without non-technical strategies, most advanced AI produced by companies like Microsoft can amplify harmful behavior. Some of the core principles to ensure a fair dataset are,

- **Intentional representation:** Deliberately ensuring diverse population coverage[3]
- **Systematic bias monitoring:** Continuous assessment during data collection rather than post-hoc detection[29]
- **Transparent methodology:** Clear documentation of collection decisions and potential limitations[16]
- **Stakeholder involvement:** Including affected communities in the collection design and validation [34]

From the start of data collection to the final dataset, bias assessment is always integrated into the process of bias-aware data collection. This methodology places bias detection and mitigation as fundamental design concepts that direct collecting decisions from the beginning, in contrast to the traditional approach, which addresses bias as an afterthought.

The scope covers all types of systematic error, such as temporal skews, geographic coverage gaps, measurement inconsistencies, and demographic under-representation, that can skew data representativeness. This method acknowledges that gathering data is not an impartial process and that many choices made during the process have the potential to introduce or reinforce preexisting biases in society. The approach addresses intricate intersectional biases and contextual factors that affect data quality and fairness, going beyond straightforward demographic quotas.

4.3.1 Traditional Approach vs Bias-Aware Approach

The traditional approach focuses more on efficiency and cost-effectiveness while collecting data. Sometimes, handling bias is left as an afterthought, and sometimes, it is not taken into account at all, which can lead to a biased prediction model. The traditional approach also assumes that population homogeneity, which is highly unlikely, and it also mostly focuses on achieving the target sample size regardless of the representativeness.

On the other hand, bias-aware data collection approach keeps this in account from the beginning till the end, resulting in a fair dataset at the end. Whereas standard methods may sample from easily available populations, bias-aware approaches specifically target underrepresented groups. While traditional approaches standardise collection procedures to ensure consistency, bias-aware methods modify procedures to accommodate different demographics and settings.

A fundamental philosophical shift is seen by the prevention versus correction paradigm in addressing bias within data collection. The correction paradigm relies on different algorithms and techniques to take care of the bias post-collection, whereas the

prevention-focused strategy includes proactive community engagement to ensure diverse participation. Along with others, some bias-aware data collection techniques are;

- **Multi-stage stratified sampling[26]** : For this method, the target population is divided into some meaningful subgroups, and these subgroups are not random; they are selected based on multiple factors, including demography, geography, or behavioural characteristics.
- **Geographic Equity Sampling[36]**: Mapping the collection sites in such a way that it is geographically accessible and carry proper socioeconomic context is the core idea of this technique, because having a collection site out of reach from a group of people can create bias in the dataset.
- **Adaptive Sampling** : This technique involves real-time monitoring for instance Computer-Assisted Telephone Interviewing(CATI)¹, so this dynamic approach monitors the collection progress and, according to the emerging pattern, the collection strategy is changed. As an example, if the data collection shows lower response from a certain demographic group, then the sampling strategy adapts by increasing outreach efforts
- **Purposive Oversampling [35]**: Intentionally oversampling groups that have been historically underrepresented or are difficult to reach in order to guarantee that their voices are heard. This method acknowledges that, because of limitations to accessibility and varying response rates, equal sample rates might not result in equal representation.
- **Trusted Intermediary Network** : To maintain a fair dataset participation of different groups, and social norms are crucial but different groups gets interested to provide data in different ways so to partnering with some established community organization, social service agency will help to design the collection strategy which is accepted and comfortable to target population.
- **Multimodal data collection** : Different data providers may face difficulties in some data collection methods, so providing multiple ways to collect data can create balance in the dataset. For example, conducting an online survey will only include people with internet access, leaving out the people without internet, so if an in-person meeting is provided along with the online survey will also include the people without internet accessibility.
- **Language and Literacy Adaptation** : Providing materials in multiple languages and at various literacy levels, including visual aids, audio options, and

¹<https://www.retently.com/blog/cati-survey/> (Accessed: June 18, 2025).

simplified language versions. This goes beyond translation to include cultural adaptation of concepts and examples.

4.3.2 Challenges

Despite the necessity of fair data and bias-aware data collection being a promising solution for this, there are a few challenges for this method, including;

- **Higher Financial Requirements** : Bias-aware data collection usually costs more than the traditional data collection strategy due to the requirement of multiple sampling or extended outreach efforts, or even specialized training. These additional costs are sometimes unexplainable to the stakeholders as the cost of biased data in the long term is unseen.
- **Additional Time Requirements** : Due to the iterative nature of the bias-aware data collection, this requires more time than the traditional approach, also due to community engagement and adaptive sampling, additional time might be required to use this method.
- **Specialized Expertise Requirements** : Implementing bias-aware techniques requires interdisciplinary expertise spanning statistics, social science, community engagement, and domain-specific knowledge. Many organizations lack in-house expertise and struggle to recruit or contract specialists, particularly for niche populations or sensitive topics.
- **Quality vs Representation** : Sometimes, efforts to improve representation run counter to goals for data quality. There may be conflict between inclusivity and conventional quality measures when trying to engage hard-to-reach communities, for instance, because doing so may need accepting lower response rates, shorter survey completion times, or less controlled collection conditions.

4.3.3 Recommendation for hiring a dataset

A hiring dataset can get biased in different aspects, including but not limited to racial bias, gender bias, and accessibility bias, etc. So some recommendations for fair data collection are to,

- **Stakeholder Engagement** : Involve hiring managers, labour advocates, employee resource groups, HR specialists, and members of historically under-represented communities in the dataset creation process. By eliminating potential blind spots in conventional HR data collecting, this guarantees that collection methods cover a variety of recruiting experiences and viewpoints.

- **Multiple channel** : Collecting data from different sources eases the pressure, and makes it leaner towards a bias-free dataset. As multiple channel collection exposes more diversity and accuracy.

4.4 Research Question 05

What data collection strategies can minimize bias?

Introduction of bias-aware data collection can be a huge beneficial step towards a bias-free dataset. Even though this technique can be very costly but it can save so much in the long run.

Chapter 5

Bias Mitigation

Following upon the findings from Bias Detection Techniques, this chapter focuses on the application and methods to address both the class imbalance of target class and feature-level bias behaviors. Class imbalance and class discrimination are interconnected challenges that require a systematic approach to bias mitigation. As a result, this chapter will address and implement at first the distributional problem of imbalance instance through data-level interventions, and then class level bias of feature using fairness-driven optimization techniques supported by algorithmic adjustments. To obtain the target for the recruitment dataset, a two-phase mitigation approach was adopted.

- Phase 1: Target Class Imbalance Mitigation
- Phase 2: Feature Bias Mitigation

Splitting mitigation strategy into two phases was made to address bias source independently and effectively. As different bias-stream sources needs or requires different methods for correction, by separating mitigation strategies will ensure techniques specially designed for its nature. It will also highlights the individual contribution of each phase to overall performance of the machine learning predictions.

5.1 Phase 1: Target Class Imbalance Mitigation

An imbalance target attributes imposes significant challenge in binary classification tasks in supervised machine learning problem as it will influence both model performance and fairness evaluation. According to to author Galar, imbalanced representation of groups are one of the key reason for dataset bias[10]. In the context of recruitment dataset, the distribution of target variable 'HiringDecision' was found skewed during uni-variate feature analysis.

```

Number of observations per hiring decision:
HiringDecision
0      1035
1       465
Name: count, dtype: int64

```

Figure 5.1: Class Imbalance of Target Variable

The Figure-5.1, shows that the 'Not Hired(0)' class is doubled than 'Hired(1)' class, as a result it leads to biased model behavior, where the classifier learns to predict majority class well and neglect to learn meaningful patterns in the minority class. Imbalance target class not only affects the model performance but also creates uncertainty to detect bias in other features by skewing the model outcome and fairness metrics. Therefore, this phase is designated on correcting the target variable imbalance using different techniques as a foundational step to ensure reliable model performance. To address the skewed target class imbalance, this section explores two different techniques to mitigate class imbalance. Each technique will be evaluated based on data quality, predictive accuracy and improving fairness across different classes. The three approaches are:

- Method 1: Resampling using SMOTE
- Method 2: Class Reweighting

5.1.1 Resampling using SMOTE

Resampling techniques such as SMOTE [39] are used to address the imbalance class in a dataset. This method adjusts the data by oversampling minority class or under-sampling majority class. In the recruitment dataset, the target variable 'HiringDecison' contains imbalance class where 'Hired' class contains 31% and 'Not Hired' class contains 69% data. As a result, the model will learn biased decision boundaries and will favor the majority class, means it will predict true negative compared to true positive. This will cause machine learning model to predict discriminatory outcomes for the minority class. To address this SMOTE (Synthetic Minority Over-Sampling Techniques) was used to generate synthetic samples for minority class.

Why we need Class Resampling using SMOTE?

- As, random oversampling technique involves duplicating existing instances in the training set, there is a probability of overfitting.
- Since the recruitment dataset was generated synthetically, SMOTE was considered to introduce diversity in minority class.

Implementation requirements and results

SMOTE algorithm was implement using 'imblearn library' with default setting 'sampling_strategy='auto'' and it was only performed in training data for preventing data leakage. After applying SMOTE, the target class distribution shown in the following Table-5.1:

Class Label	Before SMOTE	After SMOTE
Hired(1)	465 samples	1035 samples
Not Hired (0)	1035 samples	1035 samples
Class Ratio	1035 : 465 = 2.2:1	1035 : 1035 = 1 : 1 (Balanced 50:50)

Table 5.1: Target Class Distribution Before and After SMOTE

After applying Oversampling in baseline model

Model Name	Accuracy	Class	Precision	Recall	F1-score
Logistic Regression	0.80	0	0.82	0.76	0.78
		1	0.77	0.83	0.80
Decision Tree	0.93	0	0.97	0.88	0.92
		1	0.89	0.98	0.93
Random Forest	0.97	0	0.98	0.96	0.96
		1	0.96	0.97	0.97
KNN	0.81	0	0.83	0.78	0.80
		1	0.79	0.83	0.81
SVM	0.90	0	0.90	0.89	0.90
		1	0.89	0.90	0.90

Figure 5.2: Class-wise model performance after Oversampling using SMOTE

Key Observation

- After oversampling techniques, noticeable improvement in Random Forest and Decision Tree were noticed for minority class 1.
- The difference between Class 0 and Class 1 was reduced by implementing resampling, indicating improved fairness in predictions in model like Decision Tree and SVM.
- As a result, Random oversampling proved to be a simple, effective mitigation techniques without no major trade-offs and ensures more reliable predictions.

5.1.2 Reweighting

Class reweighting is a cost-sensitive learning technique used to address class imbalance by assigning different weights to its classes during model training[21]. Rather than modifying the data distribution like SMOTE or Resampling , it adjust the learning algorithm by assigning higher penalties to the minority class samples.

Why we need class Reweighting?

- **Majority Class Bias:** Traditional machine learning algorithm optimize for overall accuracy but with imbalance target class dataset, model will favor and achieve high accuracy of the majority class.
- **Minority Class Neglect:** With imbalance target class(in the recruitment context the 'Hired' class) dataset, it will lead poor 'Recall' for the minority class and because of false negative prediction in recruitment context,qualified candidates will be reject also.
- **Maintaining Original Dataset:** Unlike other class mitigation strategy, reweighting doesn't introduce synthetic data, which preserves the data quality and data authenticity.

Implementation requirements and results

Reweighting method was implemented using 'scikit-learn library', which provides built-in support for cost-sensitive learning with the help of 'class_weight='balanced'" parameter. It will automatically adjust the importance of each class by calculating its frequency in the training set.The advantage of implementing Reweighting techniques is that it's keeps the original data as it is, which makes it less overfitting. However, all the supervised model doesn't support Reweighting techniques like Resampling such as KNN model doesn't support reweighting.

The table Reweighting presents class-wise performance of recruitment dataset, showcasing their influence on improving model fairness for each class.

After applying Reweighting in baseline model

Model Name	Accuracy	Class	Precision	Recall	F1-score
Logistic Regression	0.83	0	0.90	0.85	0.87
		1	0.71	0.81	0.75
Decision Tree	0.91	0	0.92	0.94	0.93
		1	0.86	0.82	0.85
Random Forest	0.93	0	0.91	0.99	0.95
		1	0.97	0.81	0.88
KNN	0.86	0	0.87	0.94	0.90
		1	0.84	0.69	0.76
SVM	0.88	0	0.91	0.90	0.91
		1	0.80	0.82	0.81

Figure 5.3: Class-wise model performance after Reweighting

Key Observation

- After Reweighting techniques applied in the dataset, noticeable improvement was noticed across all model for minority class (Hired class).
- Because of reweighting, all the models showed improved class 1 recall especially in logistic regression, SVM, which means more positive class predictions, however, for all class 1 precision decreased slightly which may be a result of a trade-off for aiming high positive prediction by allowing more false positive predictions.
- Among all the model, Random Forest and Decision Tree maintains high accuracy with balanced metrics for both classes which makes these two models more robust towards class imbalance.

5.2 Phase 2: Feature Bias Mitigation

After addressing class imbalance in the target feature, this section focuses on eliminating bias in features. Exploratory analysis, bias detection using AI tools and explainable AI helped to identify the biased attributes, especially the critical attributes within the dataset that influence the hiring decision. These analyses revealed 'EducationLevel', 'ExperienceYear' and 'RecruitmentStrategy' exhibited bias in favor of the privileged or majority group. So, it's important to address this feature-wise bias behavior and mitigate through targeted feature selection process[23]. The study used AI tools such as 'AIF360(pre-processing method)' and 'Fairlearn(in-processing method)'

techniques to ensure and promote fair treatment across all features , without compromising class balance achievement earlier. This section presents four advanced bias mitigating mechanisms specifically designed to remove and reduce bias in qualification-based and systematic biases in the recruitment process. Among these techniques , some worked on removing discrimination from the data itself, while others enforced different fairness constraints during model training to promote fairness in both privileged and non-privileged group. The methods are:

- Reweighting (AIF360)
- Disparate Impact Remover (AIF360)
- Exponentiated Gradient-Demographic Parity (Fairlearn)
- Exponentiated Gradient – Equalized Odds (Fairlearn)

5.2.1 Reweighting (AIF360)

This pre-processing techniques of AIF360, assigns instant weights before training based on the group membership[8]. Unlike the traditional reweighting technique method, this advance AIF360 techniques considered both bias feature and target class balance simultaneously. FOr instance, the lower education level(with bachelor degree) received higher weights than the privileged group(with masters and PhD).

Why we need it?

- It explicitly considers how certain group membership interacts with target variable hiring decisions, which ensures the model learns fair patterns rather than learning unfair discriminatory trends from the past.
- It also helps to maintain balance group influence without modifying original feature values.

Implementation and Result

It was implemented using the in-built AIF360 'Reweighting' algorithm and the following tables show the result for each class of target variable.

After applying Reweighting using AIF360 in baseline model

Model Name	Accuracy	Class	Precision	Recall	F1-score
Logistic Regression	0.84	0	0.85	0.93	0.89
		1	0.80	0.66	0.71
Decision Tree	0.91	0	0.91	0.90	0.91
		1	0.78	0.81	0.79
Random Forest	0.93	0	0.91	0.99	0.95
		1	0.96	0.80	0.87
KNN	0.67	0	0.87	0.94	0.90
		1	0.84	0.69	0.76
SVM	0.86	0	0.87	0.94	0.90
		1	0.84	0.68	0.75

Figure 5.4: Class-wise model performance after applying Reweighting using AIF360

Key Observation

- Random forest showed overall good accuracy for both classes with highest accuracy 93%.
- However, decision tree showed most balanced performance in terms of F1 scores.
- finally, after applying Reweighting method, logistic regression and SVM still showing bias for majority class, whereas KNN shows the poorest performance.

5.2.2 Disparate Impact Remover (AIF360)

Unintentional or indirect bias is encoded through disparate impact[13]. So to address this, pre-processing technique using AIF360 reduces the bias in feature values to eliminate discriminatory patterns while maintaining essential predictive relationships. The goal of this method is to remove disparate impact at the feature level by repairing and balancing the correlation between certain feature with the protected ones, which ensures the minimal influence of sensitive features on hiring decisions.

Why we need it?

- Feature like distance from the company, age, number of previous working companies may correlate with the protected features of the recruitment dataset. Traditional bias mitigation techniques may overlook this proxy variables, however disparate impact remover method tackles this issue at its source so that it can neutralize the influence of these features during hiring decision.

Implementation and Result

To implement disparate impact remover with AIF360 toolkit, at first the protected features need to be identified and declared. With the help of 'DisparateImpactRemover' function, all the baseline model were retrained and the the following tables show the result for each class of target variable.

After applying Disparate Impact Remover using AIF360 in baseline model

Model Name	Accuracy	Class	Precision	Recall	F1-score
Logistic Regression	0.85	0	0.87	0.93	0.90
		1	0.81	0.70	0.75
Decision Tree	0.89	0	0.91	0.92	0.93
		1	0.84	0.82	0.83
Random Forest	0.94	0	0.94	0.98	0.96
		1	0.95	0.85	0.90
KNN	0.70	0	0.74	0.86	0.80
		1	0.51	0.33	0.40
SVM	0.87	0	0.87	0.94	0.91
		1	0.85	0.68	0.76

Figure 5.5: Model performance after bias mitigation using Disparate Impact Remover

Key Observation

- Random forest showed the best overall good accuracy and performance for both classes with with strong F1 score, means it's reliability for decision making.
- After random forest, decision tree showed most balanced performance in terms of F1 scores.
- As usual, KNN didn't perform well, showing very low precision and recall.

5.2.3 Exponentiated Gradient-Demographic Parity (Fairlearn)

This in-processing bias mitigation algorithm is available in Fairlearn library[49] , where it enforce constrains during model training which ensures the balance between predictive performance and fair treatment across all groups.

Why we need it?

This method gives the opportunity to prevent the systematic exclusion of certain groups from positive hiring decision. Traditional ML models focus on improving accuracy, which can create the discriminatory situation of unequal selection rates across groups. Exponentiated Gradient-Demographic Parity ensures each group receives equal positive prediction rates, for instance, it will ensure that lower educational level group also have the equal chance of selection as higher educational level candidates.

Implementation and Result

To implement this method, Fairlearn's 'ExponentiatedGradient' with the 'DemographicParity' constraint was used and the following table provides the results of outcomes.

After applying Exponentiated Gradient-Demographic Parity using Fairlearn in baseline model

Model Name	Accuracy	Class	Precision	Recall	F1-score
Logistic Regression	0.72	0	0.78	0.85	0.81
		1	0.57	0.46	0.51
Decision Tree	0.83	0	0.93	0.81	0.87
		1	0.67	0.86	0.76
Random Forest	0.85	0	0.92	0.86	0.86
		1	0.72	0.83	0.77
KNN	0.86	0	0.87	0.94	0.90
		1	0.84	0.69	0.76
SVM	0.82	0	0.86	0.87	0.87
		1	0.72	0.69	0.71

Figure 5.6: Model performance after bias mitigation using Demographic Parity

Key Observation

- Most of the model shows improved performance on class 1 which is 'Hired' class, which means the fairness constraints are working.
- Because of fairness constraints decision boundary, the accuracy dropped, but the fairness increased between classes.

5.2.4 Exponentiated Gradient – Equalized Odds (Fairlearn)

This method also works like 'Exponentiated Gradient-Demographic Parity' by enforcing fairness equalized odd constraints[49], aiming for equality in both true positive and

false positive rates across all groups. For example, this method ensures that candidates with less experience have similar false and true positive rates like candidates with more experience to address the influence of experience -related privileged.

Why we need it?

It maintains hiring outcome quality while ensuring fair treatment by preventing both false rejection and false acceptance. For instance, qualified candidates from all group have equal chance of selection, which addresses performance-based discrimination of the dataset.

Implementation and Result

The following tables shows the baseline model results after applying Exponentiated Gradient – Equalized Odds function using Fairlearn’s ‘ExponentiatedGradient’ with the ‘EqualizedOdds’ constraint.

After applying Exponentiated Gradient-qualized Odds using Fairlearn in baseline model

Model Name	Accuracy	Class	Precision	Recall	F1-score
Logistic Regression	0.81	0	0.83	0.90	0.87
		1	0.72	0.60	0.66
Decision Tree	0.89	0	0.92	0.93	0.92
		1	0.84	0.82	0.83
Random Forest	0.94	0	0.93	0.98	0.95
		1	0.95	0.85	0.90
KNN	0.86	0	0.87	0.94	0.90
		1	0.84	0.69	0.76
SVM	0.87	0	0.85	0.97	0.92
		1	0.90	0.63	0.74

Figure 5.7: Model performance after bias mitigation using Equalized Odds

Key Observation

- Random Forest showed balance precision and recall for both of the classes, indicating successful bias mitigation.
- It shows improved error rate balance for different groups as their is reflection of balance recall and precision for all classes.

5.3 Research Question 2 and 4

Through the use of technical bias detection approaches such as exploratory data analysis, AI toolkit AIF360 and Fairlearn, the second research question was addressed. Integrating explainable AI in detection bias within dataset address the research question 04 for this study. All of this method helps to analyze bias source and the reason behind bias occurrence and why the model make certain predictions so that it can increase trust and fairness in recruitment process.

Chapter 6

Result Comparison And Analysis

This chapter provides comprehensive analysis of five machine learning algorithm across different approaches before and after applying various bias mitigation techniques. Among these approaches one is the baseline models and the rest six are bias mitigation and fairness improvement methods. For the recruitment dataset, two categories of mitigation strategies were applied and implemented:

- Target class imbalance (Using oversampling with SMOTE and Reweighting)
- Feature-level bias mitigation (Reweighting (AIF360), Disparate Impact Remover (AIF360), Exponentiated Gradient with Demographic Parity and Equalized Odds (Fairlearn)).

For the initial assessment, each model ‘Hired (1) and ‘Not Hired (0) class outcomes were evaluated based on Accuracy, precision, Recall and F1-score.

6.1 Baseline Performance

At first, all five ML models were implemented using raw dataset without bias mitigation. All the models show high accuracy predicting ‘Not Hired (0)’ class, whereas lower for ‘Hired’ class. This indicates dataset imbalance or bias in the feature.

The following Table-6.1, shows a compact comparison of the baseline model prediction across both classes .

Model	Accuracy	F1-score(class 0)	F1-score (class 1)
Logistic Regression	0.87	0.91	0.76
Decision Tree	0.88	0.91	0.80
Random Forest	0.93	0.95	0.87
KNN	0.89	0.93	0.78
SVM	0.90	0.93	0.81

Table 6.1: Performance Comparison for baseline model

Best baseline Model: Random Forest with 93% accuracy with F1-score for class 1: 0.95 and for class 0: 0.87

6.2 Target Class Imbalance mitigation

To balance and improve the results of machine learning model, 2 traditional dataset bias mitigation techniques were implemented where it showed that applying SMOTE improves the result of Random Forest.

Traditional Reweighting

Traditional reweighting aims to balance the representation of different classes of target variable within the dataset to improve fairness during model training, shown in Table-6.2.

Model	Accuracy	F1-score(class 0)	F1-score (class 1)
Logistic Regression	0.83	0.87	0.75
Decision Tree	0.91	0.93	0.85
Random Forest	0.93	0.95	0.88
KNN	0.86	0.90	0.76
SVM	0.88	0.91	0.81

Table 6.2: Performance comparison after Reweighting

Traditional Oversampling using SMOTE

Traditional oversampling using SMOTE is a data-level bias mitigation techniques and aims to balance the representation of different classes by generating synthetic samples to improve fairness during model training, shown in Table-6.3.

Model	Accuracy	F1-score(class 0)	F1-score (class 1)
Logistic Regression	0.80	0.78	0.80
Decision Tree	0.93	0.92	0.93
Random Forest	0.97	0.96	0.97
KNN	0.81	0.80	0.81
SVM	0.90	0.90	0.90

Table 6.3: Performance comparison after Oversampling using SMOTE

Comparison of Baseline model with Traditional bias mitigation techniques

Table 6.4 compares the F1-scores of Reweighting and oversampling with SMOTE techniques for both classes against baseline model. The Δ values shows the changes relative to the baseline and highlights the performance after target class imbalance mitigation.

Model	Baseline		SMOTE		Reweighting	
	C0	C1	Δ C0	Δ C1	Δ C0	Δ C1
Logistic Regression	0.91	0.76	-0.13	+0.04	-0.04	-0.01
Decision Tree	0.91	0.80	+0.01	+0.13	+0.02	+0.05
Random Forest	0.95	0.87	+0.01	+0.10	0.00	+0.01
KNN	0.93	0.78	-0.13	+0.03	-0.03	-0.02
SVM	0.93	0.81	-0.03	+0.09	-0.02	0.00

Table 6.4: F1-score and Δ Change for Both Classes after SMOTE and Reweighting

Results

- From the Table-6.4, it is evident that decision tree and random forest resulted in positive F1-score in both SMOTE and reweighting techniques.
- However, the changes of Class 1 (+0.10) for random forest in SMOTE technique is the highest improvement.
- It shows that random forest is the most consistent and robust model , as it achieved accuracy(Baseline-94% and after SMOTE-96%) and fairness without any loss of performance.

6.3 Feature-level bias mitigation

Comparison of Baseline model with AIF360 techniques

The following Table 6.5 shows the performance changes after mitigating bias in feature level with the help of AIF360.

Model	Baseline		AIF360-DIR		AIF360-Reweighting	
	C0	C1	Δ C0	Δ C1	Δ C0	Δ C1
Logistic Regression	0.91	0.76	-0.01	-0.01	-0.02	-0.05
Decision Tree	0.91	0.80	+0.02	+0.03	0.00	-0.01
Random Forest	0.95	0.87	+0.01	+0.03	0.00	0.00
KNN	0.93	0.78	-0.13	-0.38	-0.03	-0.02
SVM	0.93	0.81	-0.02	-0.05	-0.03	-0.06

Table 6.5: F1-score and Δ Change for Both Classes after AIF360 mitigation techniques

Results

- From the Table-6.4, AIF360's DIR improved Class 1 F1-scores in decision tree and random forest (+0.03 each), but led to performance drops in models like Logistic Regression and SVM and drastically for KNN.
- On the other hand , Reweighting produced more stable results compared to DIR as the changes in F1-score is minimal. The outcomes in random forest remains unchanged which is the best scenarios in reweighting compare to other results.

Comparison of Baseline model with Fairlearn techniques

The following Table 6.6 shows the performance changes after mitigating bias in feature level with the help of Fairlearn.

Model	Baseline		Fairlearn-DP		Fairlearn-EO	
	C0	C1	Δ C0	Δ C1	Δ C0	Δ C1
Logistic Regression	0.91	0.76	-0.10	-0.25	-0.04	-0.10
Decision Tree	0.91	0.80	-0.04	-0.04	+0.01	+0.03
Random Forest	0.95	0.87	-0.09	-0.10	0.00	+0.03
KNN	0.93	0.78	-0.03	-0.02	-0.03	-0.02
SVM	0.93	0.81	-0.06	-0.10	-0.01	-0.07

Table 6.6: F1-score and Δ Change for Both Classes after Fairlearn mitigation techniques

Results

- After implementing exponentiated Gradient-Demographic Parity, the F1-score of all models gets decreased specially models like random forest, linear regression and SVM.
- On the other hand , after applying exponentiated Gradient – Equalized Odds, it showed both decision tree and random forest performs well compare to exponentiated Gradient-Demographic Parity, as both the models gained in class 1 by +0.03.

6.4 Comparison between performance

6.4.1 Research Question 03

Across multiple mitigation strategies, decision tree performed well for both of the classes compare to its Δ values. This suggests that decision tree was responsive to fairness aware techniques. However, in terms of overall accuracy random forest constantly maintained high baseline and post mitigation performance and accuracy.

This highlights, a trade-off between predictive models accuracy and fairness: while random forest remains most reliable throughout the study. The following Table-6.7, shows the most successful models during different mitigation strategies based on accuracy.

Method	Best Model	Accuracy	F1 (Class 0)	F1 (Class 1)	Comments
Baseline	Random Forest	93%	0.95	0.87	Excellent
SMOTE Oversampling	Random Forest	97%	0.96	0.97	Outstanding
Manual Reweighting	Random Forest	93%	0.95	0.88	Excellent
AIF360 Reweighting	Random Forest	93%	0.95	0.87	Excellent
Disparate Impact Remover	Random Forest	94%	0.96	0.90	Excellent
EG - Demographic Parity	KNN	86%	0.90	0.76	Good
EG - Equalized Odds	Random Forest	94%	0.95	0.90	Excellent

Table 6.7: Performance Summary of Best Models per Mitigation Method

It showed that in every method random Forest performed well not only in achieving high accuracy but also maintaining fairness between target classes.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

This thesis explored the source of bias within dataset and analysed the impact of bias in recruitment datasets and investigated a range of technical methods for detecting and mitigating such bias to ensure fair and accurate AI-driven hiring decisions. The study began with a synthetic recruitment dataset consisting of various candidate features including age, gender, education level, years of experience, number of previous companies worked, interview score, skill score, personality score, recruitment strategy, and the hiring decision.

The research began by identifying key sources of bias within recruitment dataset using exploratory data analysis and fairness metrics revealed that features such as **EducationLevel**, **ExperienceYears**, **RecruitmentStrategy** were contributing to unequal outcomes. It is worth mentioning that, the study also revealed that the traditional bias source such as **Age and Gender** had no significant negative influence in this dataset. These findings confirmed the presence of both direct and indirect discrimination embedded mostly with the structural factors of hiring for instance, education level, experience years and recruitment strategy.

This study investigates the bias features and their impact thoroughly. To address this, at first five commonly used classification models were implemented and treated as the baseline model. Across all baseline models evaluation showed that, predictions for Class 1 (Hired) were consistently weaker than those for Class 0 (Not Hired).

As a result, a series of bias detection and mitigation methods were implemented to compare with baseline model using tools such as AIF360 and Fairlearn, applying strategies like Reweighting, Disparate Impact Remover, and Exponentiated Gradient (with

both Demographic Parity and Equalized Odds constraints). In addition, Explainable AI (XAI) tools such as LIME and SHAP were used to interpret model behavior and highlight feature influence, reinforcing the transparency and accountability of the models. By doing all these process, it showed that Random Forest model performs well with raw dataset and with bias mitigated dataset as well in terms of accuracy, where as decision tree worked well to improve its prediction for Class 1('Hired').

Beyond the technical approach to address bias within dataset, this study also emphasized the importance of bias-aware data collection strategies for fair data collection. In conclusion, this thesis highlights that bias in AI systems can not overcome completely, but a challenge that can be addressed through intentional design, thoughtful data practices, and continuous evaluation because no single methods or AI tool can guarantee complete fairness.

7.2 Limitations

Despite the comprehensive approach of this study to detect and mitigate bias in recruitment dataset using machine learning model, there are several limitation that must be acknowledge. This limitations related to dataset explanation, data quality etc.

- **Synthetic Data:** The recruitment dataset used in this study was synthetically generated, as a result it may not capture the full complexity and understanding of real-world recruitment data.
- **Missing Contextual Information:** Some important features—such as candidate hiring strategy channels are not described properly, as the human and contextual meaning can be varied in different scenarios it was difficult to quantify the influence of this features in the hiring decision.
- **Limited feature:** Some potentially influential features like religion, ethnicity, educational institution name were missing as a result it limited the scope of analysing bias in real-world scenarios.

7.3 Future Work

- **Real-World Dataset Validation:** More future research should perform and validate the results on real-world recruitment datasets. This would give the opportunity for a deeper insight into practical fairness challenges of recruitment process.

- **Incorporating more feature**

Future research should integrate more contextual features for example hiring manager review, assessment score interpretation into the dataset. These features are crucial in real-world recruitment process scenarios so combining this feature will improve fairness and model interpretability.

- **Explainability in Practice**

Further research could test how integration of explainable AI tools like LIME and SHAP are interpreted by HR practitioners and whether they actually increase trust and understanding in AI decisions.

- **Generalizing Across Domains and Datasets**

Though the study only focused on recruitment dataset, further work can test the approach of bias detection and mitigation in different sectors such as loan approvals and healthcare.

- **Bias Prevention in Data Collection Design**

Finally, though the study only focused and discussed the data collection strategies conceptually, future work could involve designing and testing real-world data collection frameworks aimed at minimizing bias from the dataset.

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